PANCHIP

PAN3029 / PAN3060 系列

产品说明书

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低功耗远距离无线收发芯片

Overview

PAN3029 / PAN3060 is a low-power long-distance wireless transceiver chip using ChirpIoTTM modulation and demodulation technology. It supports half-duplex wireless communication and operates in the frequency bands of 408~565MHz and 816~1080MHz. The chip has the characteristics of high anti-interference, high sensitivity, low power consumption and ultra-long transmission distance. It has a maximum sensitivity of -143dBm and a maximum output power of 20dBm, which produces an industry-leading link budget, making it the best choice for long-distance transmission and applications with extremely high reliability requirements. Compared with conventional modulation technology, PAN3029 / PAN3060 also has significant advantages in blocking and adjacent channel selection, which can further improve communication reliability. At the same time, it also provides greater flexibility. Users can adjust the spread spectrum modulation bandwidth, spreading factor and error correction rate by themselves, effectively improving the trade-off between distance, anti-interference ability and power consumption of chips using conventional modulation technology.

Key Features

- Frequency: 408~565MHz, 816~1080MHz
- Modulation: ChirpIoT™
- Transmit power:-30dBm~20dBm
- Maximum link budget: 163dB
- Sensitivity: -143dBm@62.5kHz
- Current consumption:
  - Sleep current: 200nA
  - Receiving current:4.1mA@DCDC mode
  - Transmitting current:
    - 91mA@20dBm
    - 13mA@-2dBm
- Bandwidth
  - PAN3029：62.5kHz、125kHz、250kHz、500kHz
  - PAN3060：125kHz、250kHz、500kHz
- SF factor and automatic identification of spreading factor
  - PAN3029：5~12
  - PAN3060：5~9
- Coding rate: 4/5，4/6，4/7，4/8
- CAD function
- Data rate: 0.08~59.9kbps
- 4-wire SPI/3-wire SPI/I2C configuration interface, 4 GPIOs
- Operating Voltage: 1.8V ~3.6V (DCDC mode 2V~3.6V)
- Working temperature: -40℃~85℃
- Package: QFN28（4×4mm）

Typical Applications

- Smart Factory
- Smart Water
- Smart Agriculture
- Smart Healthcare
- Smart Community
- Smart Fire Protection

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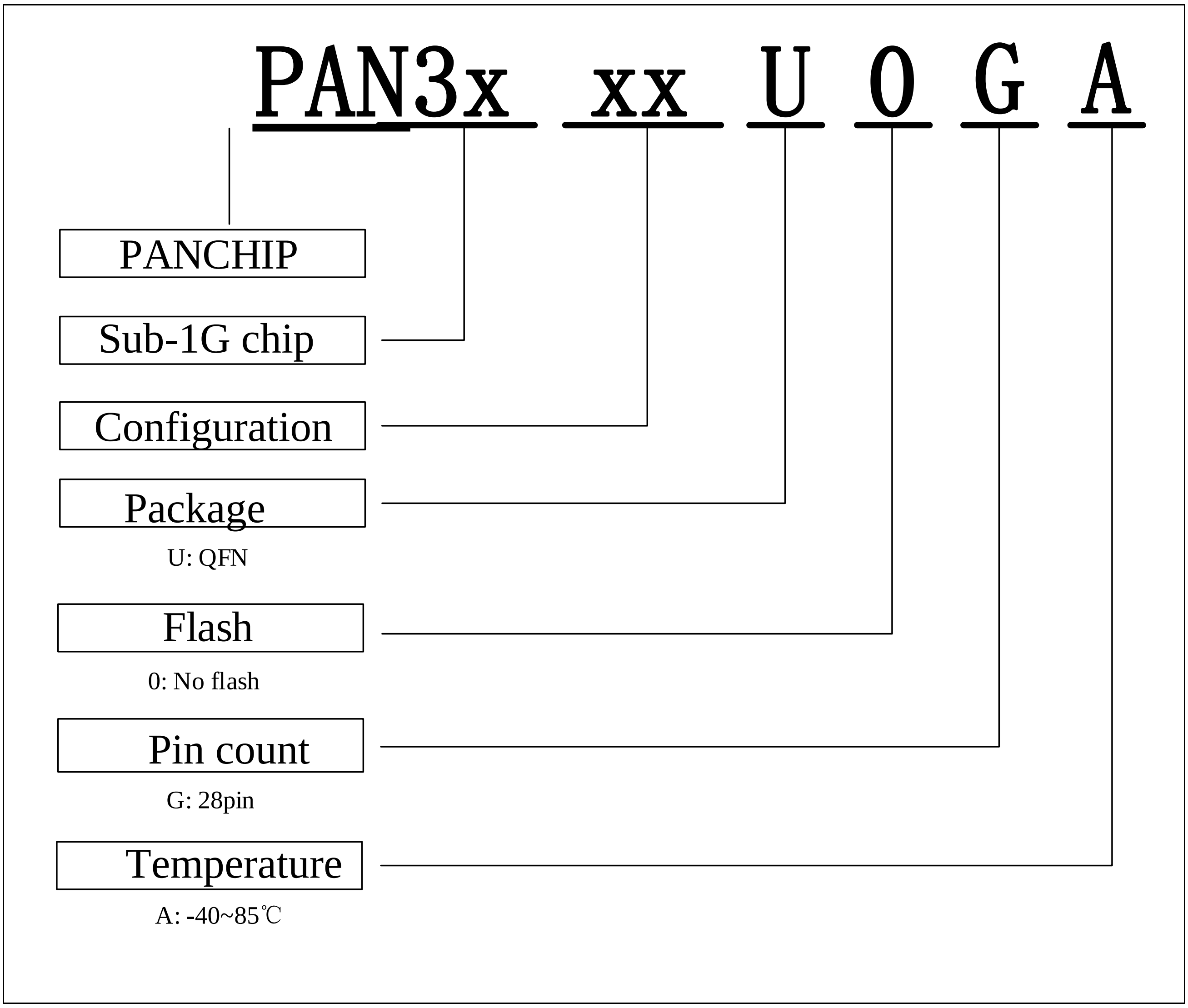
1 Naming convention



2 Ordering Information

Model	Chip Type	Package	Pin count	IO	BW	SF	Bit rate	Temperature	Package
PAN3029 U0GA	Sub-1G	QFN	28	4	62.5kHz 125kHz 250kHz 500kHz	SF5~SF12	0.08~ 59.9kbps	-40~85℃	Tape & Reel
PAN3060 U0GA	Sub-1G	QFN	28	4	125kHz 250kHz 500kHz	SF5~SF9	0.5~ 59.9kbps	-40~85℃	Tape & Reel

Before ordering, please consult sales for the latest mass production information.

This manual describes the maximum package and the most complete functions. For the configuration differences of each model, please refer to the ordering information on this page.

3 System structure block diagram

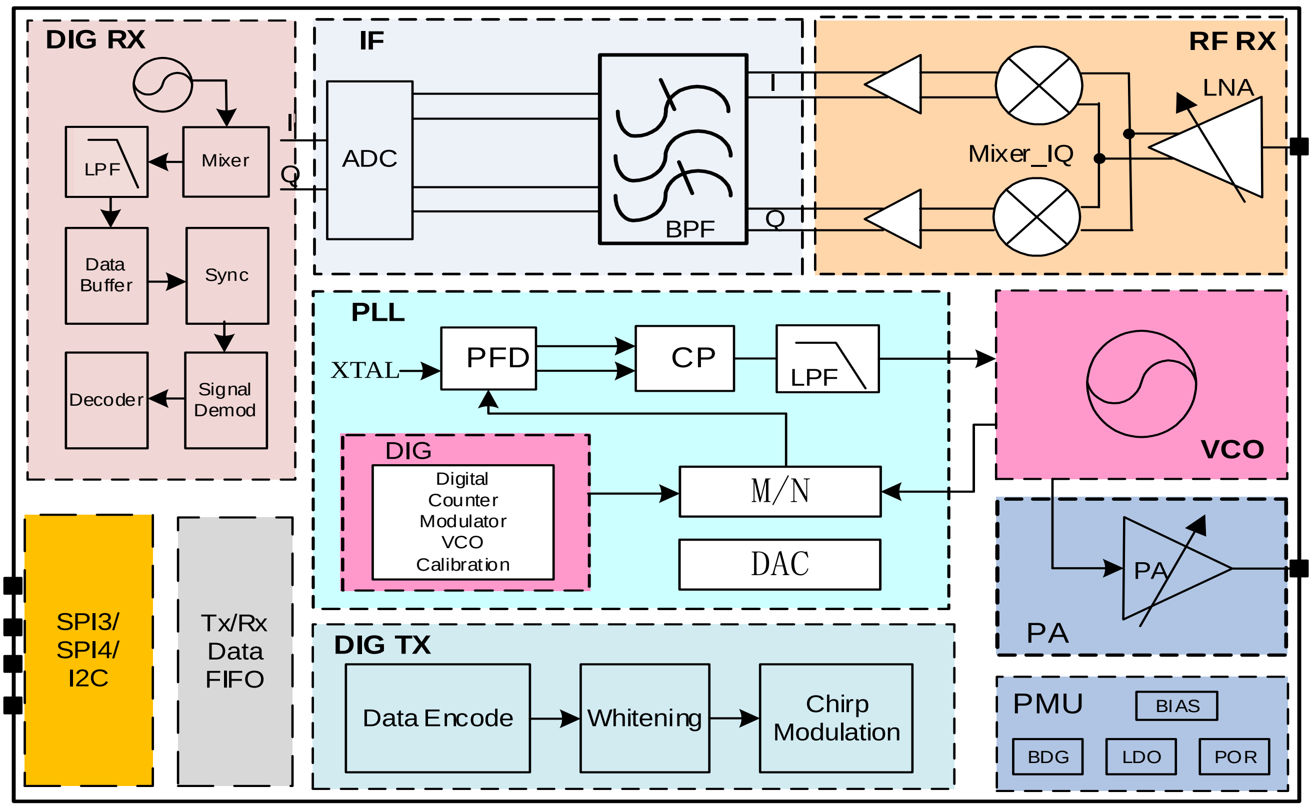


Figure 3-1 System Block Diagram

4 Pin Definition and Description

4.1 Pin Diagram

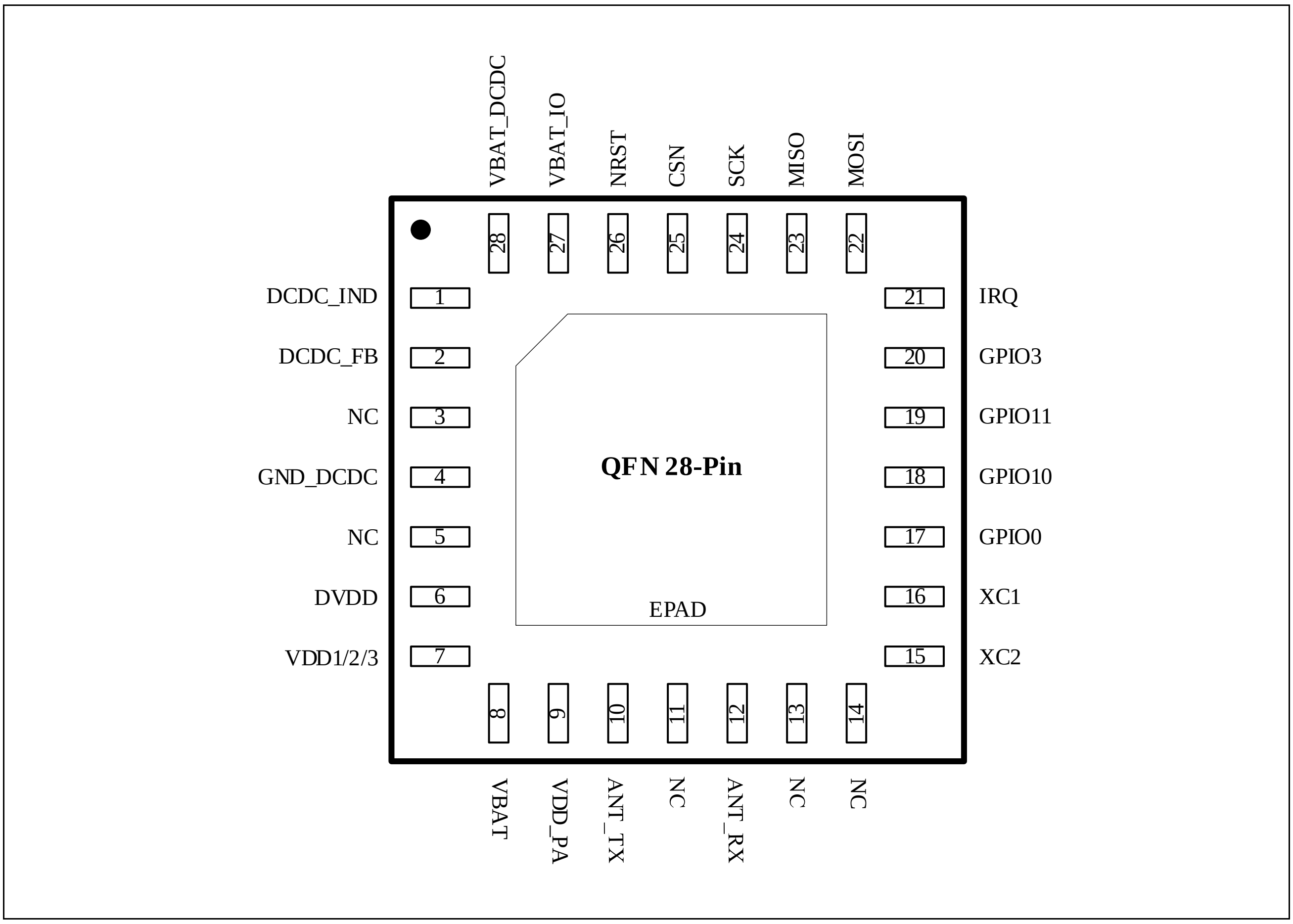


Figure 4-1 Pin diagram

4.2 Pin Description

Table 4-1 Pin Description

Pin No.	Pin Name	Type	Description
1	DCDC_IND	AO	Internal DCDC output, connected to external series inductor
2	DCDC_FB	AI	Internal DCDC feedback input, DCDC mode connected to VDD1/2/3
3	NC		
4	GND_DCDC	G	Analog Ground
5	NC		
6	DVDD	P	Digital Power LDO Output
7	VDD1/2/3	P	Analog power supply, DCDC mode connects to VFB, non-DCDC mode connects to main power supply
8	VBAT	P	Analog power supply, connected to the main power supply
9	VDD_PA	P	PA power supply LDO output
10	ANT_TX	AO	Transmitter PA Output
11	NC		
12	ANT_RX	AI	Receiver LNA Input
13	NC		
14	NC		
15	XC2	AI	Crystal input
16	XC1	AO	Crystal output
17	GPIO0	I/O	Digital IO
18	GPIO10	I/O	Digital IO
		O	External PA control signal
19	GPIO11	I/O	Digital IO
		O	Channel status indicator signal
20	GPIO3	I/O	Digital IO
21	IRQ	O	Interrupt signal
22	MOSI	I	SPI Data input signal
23	MISO	O	SPI Data output signal
24	SCK	I	SPI Serial Clock
25	CSN	I	SPI Chip select signal
26	NRST	I	Reset
27	VBAT_IO	P	Digital GPIO power supply, connected to the main power supply
28	VBAT_DCDC	P	DCDC power supply, connected to the main power supply

29	E-PAD	G	GND pad at the bottom of the chip, needs to be grounded

5 Electrical Characteristics

All parameters in this section are based on TA=25℃, power supply voltage 3.3V test, unless otherwise specified.

5.1 Absolute Maximum Ratings

Table 5-1 Absolute Maximum Ratings

Parameter	Description	Condition	Min	Typical	Max	Unit
VDD	VDD1/VDD2/VDD3/VBAT/VBAT_IO	Freq: 490MHz	-0.3	3.3	3.6	V
V_I	Input voltage		-0.3	-	VDD	V
V_O	Output voltage		VSS	-	VDD	V
T_{OP}	Operating temperature		-40	-	85	℃
T_{STG}	Storage temperature		-55	-	125	℃

Note: Exceeding one or more limit values may cause permanent damage to the chip. Electrostatically sensitive device, comply with protection rules when handling.

5.2 DC characteristics

Test conditions:

- Frequency: 490 MHz

Table 5-2 Voltage and current (frequency 490MHz)

Parameter	Description	Min	Typical	Max	Unit	Condition
VDD	Power supply voltage	1.8	3.3	3.6	V	T_A=25℃，LDO mode
		2	3.3	3.6	V	T_A=25℃，DCDC mode
VSS	Ground voltage	-	0	-	V	-
$I_{DeepSleep}$	Deep sleep current	-	200	-	nA	-
$I_{TX,20dBm}$	TX current	-	91	-	mA	20dBm Output Power
$I_{TX,18dBm}$	TX current	-	77	-	mA	18dBm Output Power
$I_{TX,-2dBm}$	TX current	-	13	-	mA	-2dBm Output Power
$I_{RX,LDO}$	RX current, LDO	-	7.8	-	mA	LDO mode, maximum LNA gain
$I_{RX,DCDC}$	RX current, DCDC	-	4.1	-	mA	DCDC mode, maximum LNA gain
V_{OH}	Output high level	VDD-0.3	-	VDD	V	-
V_{OL}	Output low level	VSS	-	VSS+0.3	V	-
V_{IH}	Input high level	0.8*VDD	-	-	V	-

V_{IL}	Input low level	-	-	0.2*VDD	V	-
SPI_rate	SPI bit rate	-	-	10	Mbps	-

Test conditions:

Frequency: 915MHz

Table 5-3 Voltage and current (frequency 915MHz)

Parameter	Description	Min	Typical	Max	Unit	Condition
VDD	Power supply voltage	1.8	3.3	3.6	V	T_A=25℃，LDO mode
		2	3.3	3.6	V	T_A=25℃，DCDC mode
VSS	Ground voltage	-	0	-	V	-
$I_{DeepSleep}$	Deep sleep current	-	200	-	nA	-
$I_{TX,20dBm}$	TX current	-	100	-	mA	20dBm Output Power
$I_{TX,18dBm}$	TX current	-	81	-	mA	18dBm Output Power
$I_{TX,-2dBm}$	TX current	-	15	-	mA	-2dBm Output Power
$I_{RX,LDO}$	RX current, LDO	-	7.8	-	mA	LDO mode, maximum LNA gain
$I_{RX,DCDC}$	RX current, DCDC	-	4.1	-	mA	DCDC mode, maximum LNA gain
V_{OH}	Output high level	VDD-0.3	-	VDD	V	-
V_{OL}	Output low level	VSS	-	VSS+0.3	V	-
V_{IH}	Input high level	0.8*VDD	-	-	V	-
$V_{I}L$	Input low level	-	-	0.2*VDD	V	-
SPI_rate	SPI bit rate	-	-	10	Mbps	-

5.3 RF Characteristics

Test conditions:

- Frequency: 490MHz
- Error Correction Code = 4/8
- Packet error rate ≤ 5%
- Payload length = 10 Bytes

Table 5-4 General RF characteristics (frequency 490MHz)

Parameter	Description	Condition	Min.	Typical	Max.	Unit
General parameters						
F_{op}	Operating frequency	-	408	-	565	MHz
		-	816	-	1080	MHz
F_{xtal}	Crystal frequency	-	-	32	-	MHz
R_S	Crystal series resistance	-	-	30	50	Ω
C_{FOOT}	Crystal external capacitance	-	8	15	22	pF
C_{LOAD}	Crystal load capacitance	-	6	10	12	pF
F_{TOL}	Initial frequency tolerance	-	-	±10	-	ppm
BR	Bit rate (PAN3029 series)	-	0.08	-	59.9	kbps
	Bit rate (PAN3060 series)	-	0.5	-	59.9	kbps
Transmitter						
P_{LPWAN}	Output Power	-	-30	-	20	dBm
Receiver						
RF_62.5（LDO）	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX path 62.5 kHz bandwidth	SF = 7 SF = 10 SF = 12	- - -	-129 -137 -143	- - -	dBm
RF_125（LDO）	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX path 125 kHz bandwidth	SF = 7 SF = 10 SF = 12	- - -	-126 -134 -140	- - -	dBm
RF_250（LDO）	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX path 250 kHz bandwidth	SF = 7 SF = 10 SF = 12	- - -	-124 -132 -137	- - -	dBm
RF_500（LDO）	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX path 500 kHz bandwidth	SF = 7 SF = 10 SF = 12	- - -	-121 -129 -134	- - -	dBm
RF_62.5（DCDC）	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX path 62.5 kHz bandwidth	SF = 7 SF = 10 SF = 12	- - -	-125 -133 -139	- - -	dBm
RF_125（DCDC）	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX path 125 kHz bandwidth	SF = 7 SF = 10 SF = 12	- - -	-122 -130 -137	- - -	dBm
RF_250（DCDC）	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX path 250 kHz bandwidth	SF = 7 SF = 10 SF = 12	- - -	-120 -128 -134	- - -	dBm

RF_500 （DCDC）	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX path 500 kHz bandwidth	SF = 7 SF = 10 SF = 12	- - -	-117 -125 -130	- - -	dBm

Test conditions:

- Frequency: 915MHz
- Error Correction Code = 4/8
- Packet error rate ≤ 5%
- Payload length = 10 Bytes

Table 5-5 General RF characteristics (frequency 915MHz)

Parameter	Description	Condition	Min.	Typical	Max.	Unit
General parameters						
F_{op}	Operating frequency	-	408	-	565	MHz
		-	816	-	1080	MHz
F_{xtal}	Crystal frequency	-	-	32	-	MHz
R_S	Crystal series resistance	-	-	30	50	Ω
C_{FOOT}	Crystal external capacitance	-	8	15	22	pF
C_{LOAD}	Crystal load capacitance	-	6	10	12	pF
F_{TOL}	Initial frequency tolerance	-	-	±10	-	ppm
BR	Bit rate (PAN3029 series)	-	0.08	-	59.9	kbps
	Bit rate (PAN3060 series)	-	0.5	-	59.9	kbps
Transmitter						
P_{LPWAN}	Output Power	-	-30	-	20	dBm
Receiver						
RF_62.5 （LDO）	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX path 62.5 kHz bandwidth	SF = 7 SF = 10 SF = 12	- - -	-128 -136 -141	- - -	dBm
RF_125 （LDO）	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX path 125 kHz bandwidth	SF = 7 SF = 10 SF = 12	- - -	-125 -132 -138	- - -	dBm
RF_250 （LDO）	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX path 250kHz bandwidth	SF = 7 SF = 10 SF = 12	- - -	-121 -129 -135	- - -	dBm
RF_500 （LDO）	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX path 500 kHz bandwidth	SF = 7 SF = 10 SF = 12	- - -	-119 -127 -132	- - -	dBm
RF_62.5 （DCDC）	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX path 62.5 kHz bandwidth	SF = 7 SF = 10 SF = 12	- - -	-128 -136 -140	- - -	dBm
RF_125 （DCDC）	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX path 125 kHz bandwidth	SF = 7 SF = 10 SF = 12	- - -	-125 -132 -137	- - -	dBm
RF_250	RF sensitivity, long range mode, maximum	SF = 7	-	-120	-	dBm

（DCDC）	LNA gain, using separate RX/TX path 250 kHz bandwidth	SF = 10 SF = 12	- -	-129 -135	- -	
RF_500 （DCDC）	RF sensitivity, long range mode, maximum LNA gain, using separate RX/TX path 500 kHz bandwidth	SF = 7 SF = 10 SF = 12	- - -	-118 -126 -130	- - -	dBm

6 Frequency synthesizer

The chip integrates high-performance RF transceiver circuits and can support multiple Sub-1GHz bands such as 400M, 800M, etc.

The chip contains a high-performance, low phase noise, fully integrated fractional frequency synthesizer. The frequency synthesizer is used to generate the LO frequency used by the down-conversion mixer in the receive mode. In the transmit mode, it is also used to directly generate the modulated RF carrier. The fractional frequency synthesizer provides excellent phase noise performance, frequency resolution less than 100Hz, and has low power consumption. The frequency synthesizer has a fast frequency stabilization function, which can complete the receive and transmit frequency stabilization in a very short time to optimize the chip power consumption.

7 Modem

The chip uses a modem that supports ChirpIoT™ transmission, and achieves an ultra-high communication link budget through excellent spread spectrum modulation technology and forward error correction technology. Compared with the traditional FSK-based modulation method, the range and stability of the radio communication link are increased.

The ChirpIoT™ modem has greatly enhanced its anti-interference ability. Compared with the FSK modulation method, the ChirpIoT™ modem can suppress co-channel interference by up to 19dB, that is, the interference signal energy can be 19dB higher than the ChirpIoT™ signal, which greatly ensures that the ChirpIoT™ signal can simply coexist with other modulation systems and is widely used in hybrid network communications.

7.1 Modulation parameters

The chip optimizes the transmission performance of ChirpIoT™ signals by configuring different modulation parameters to achieve a balance between transmission rate, transmission distance, and transmission reliability. The specific parameters are as follows:

- Signal bandwidth (BW)
- Spreading Factor (SF)
- Coding rate (CR)
- Low rate mode (LDR)

7.1.1 Signal bandwidth

PAN3029 supports four signal bandwidths: 500kHz, 250kHz, 125kHz, and 62.5kHz. PAN3060 supports three signal bandwidths: 500kHz, 250kHz, and 125kHz.
Different signal bandwidths can be selected through register configuration.

7.1.2 Spreading Factor

PAN3029 can support 8 different spreading factors from SF5 to SF12, and PAN3060 can support 5 different spreading factors from SF5 to SF9. The larger the spreading factor, the lower the data transmission payload rate.

7.1.3 Coding Rate

The chip uses forward error correction technology and supports four different coding rates: CR45, CR46, CR47, and CR48. This forward error correction technology can correct error bits within a certain range and improve transmission reliability.

7.1.4 Low rate mode

The low-rate mode is a communication mode of ChirpIoT™ modulation technology, which improves the reliability of signal transmission by reducing the effective bit information carried per unit time.

7.2 Frame structure

The frame structure of ChirpIoT™ transmission signal consists of four parts: signal Preamble, Header, Payload, and CRC, as shown in the following figure.

Preamble	Header	CRC	Payload	Payload CRC

Figure 7-1 PHY frame structure

8 MAC Design

8.1 Transmit and receive modes

The chip's transmitter has two separate operating modes, and the receiver has three separate operating modes.

Transmitter operating modes:

- Single packet mode
- Continuous mode

Receiver operating modes:

- Single packet mode
- Single packet mode with timeout
- Continuous mode

8.2 Transmitter Mode

In the MAC transmission mode, there are two modes: single packet transmission and continuous transmission. The available mode selection register can be used for selection. After the mode selection register selects the TX mode, the state machine enters the TX preparation state, waits for the FIFO to be filled with data, and then enters the TX transmission state.

When the MAC enters the TX transmission mode, the power consumption control module successively turns on the LDO module, PLL module, PA and other analog module circuits, and then sends the TX start signal to the Modem module to start sending data.

After the data is sent, the LDO module, PLL module, PA and other analog module circuits are turned off in turn. Then the IRQ signal is sent to the MCU. After the MCU clears the IRQ, the REG_OPT_MODE register can be configured to enter the STB3 mode and end a TX process.

8.2.1 Single packet mode

The state flow chart of the single packet sending mode is shown in the figure 8-1.

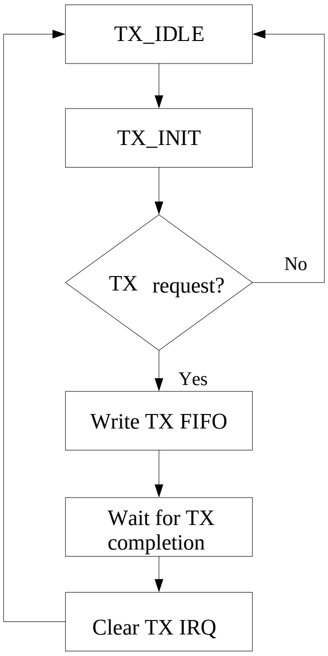


Figure 8-1 Single Packet Transmission Mode State Flowchart

8.2.2 Continuous mode

The state flow chart of the continuous transmission mode is shown in Figure 8-2.

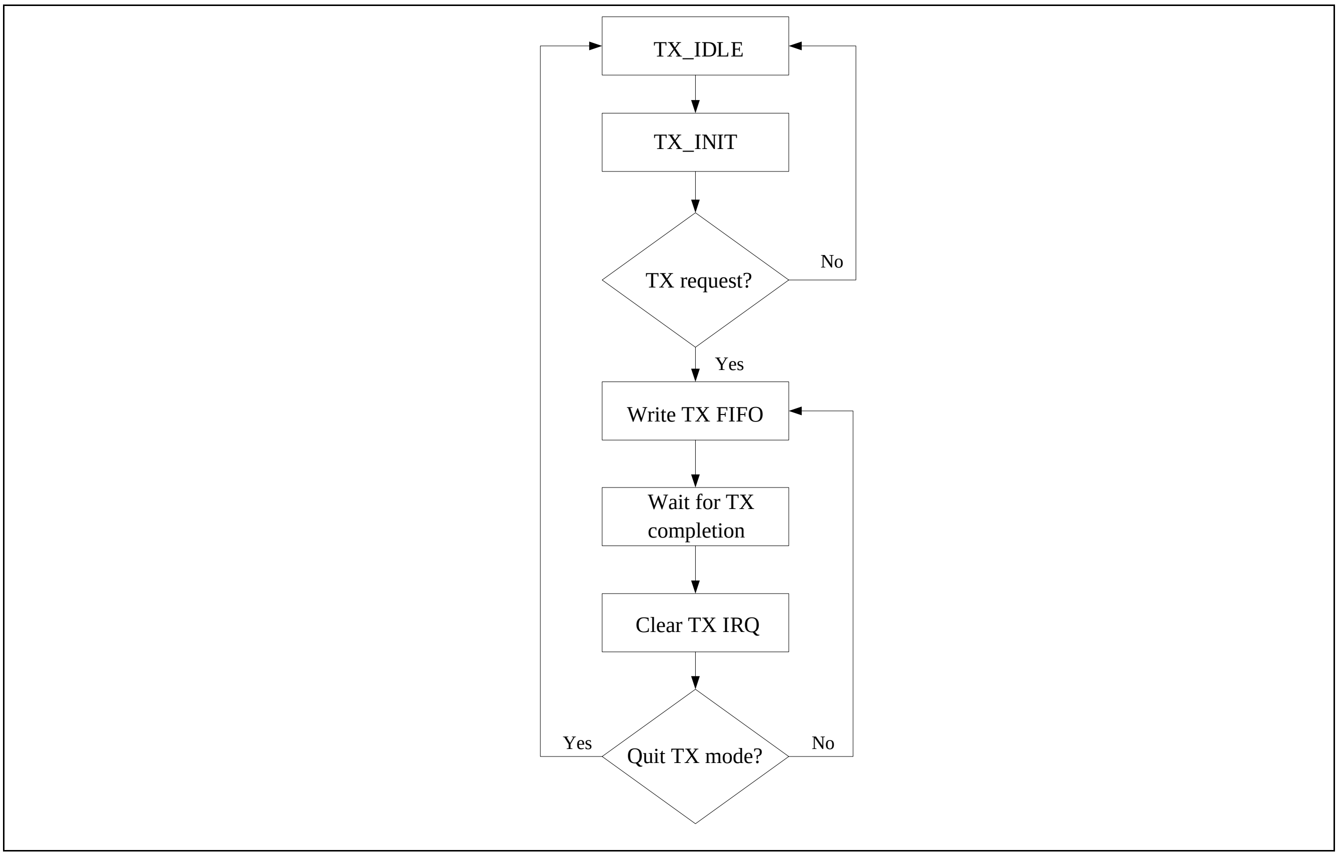


Figure 8-2 Continuous Transmission Mode Flowchart

8.3 Receiver Mode

In the MAC receiving mode, there are single packet receiving, single packet receiving with timeout, and continuous receiving modes, which are selected using a 2-bit mode selection register.

After the mode selection register selects the RX mode, the state machine enters the RX receiving state.

When the MAC enters the receiving mode, the power control module successively turns on analog module circuits such as LDO and PLL, and sends an RX start signal to the Modem to start receiving data.

After the data is received, analog module circuits such as LDO and PLL are turned off in turn. Then an IRQ signal is sent to the MCU. After the MCU clears the IRQ, the REG_OPT_MODE register can be configured to enter the STB3 mode and end an RX process.

8.3.1 Single packet mode

The state flow chart of the single packet receiving mode is shown in Figure 8-3.

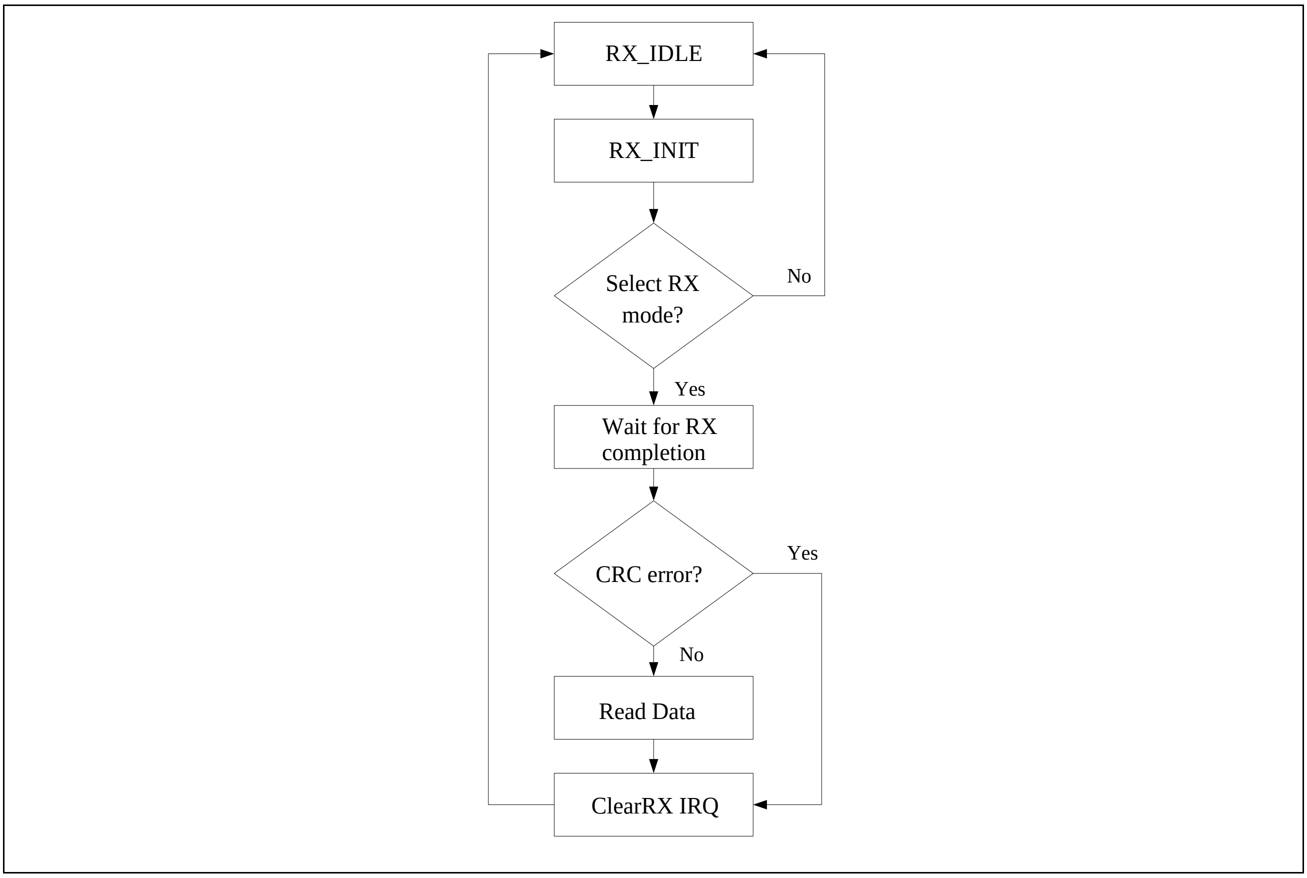


Figure 8-3 Single Packet Reception Mode Flowchart

8.3.2 Single packet mode with timeout

The state flow diagram of the single-packet receive mode with timeout is shown in Figure 8-4.

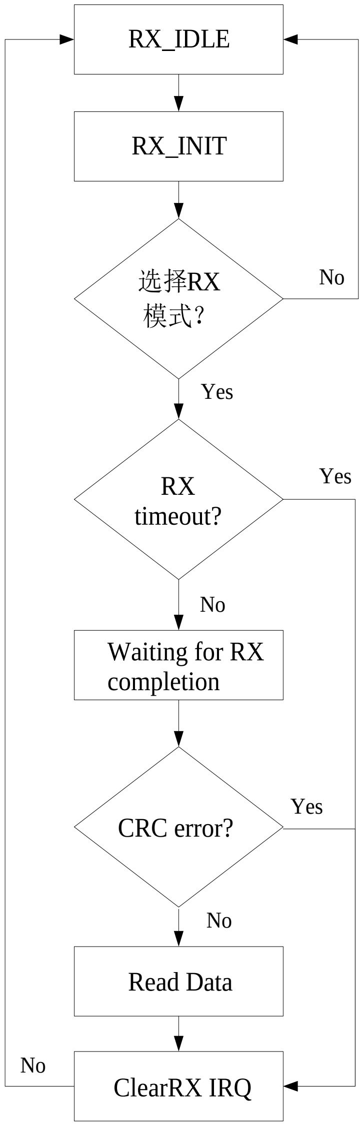


Figure 8-4 Single Packet Reception Mode Flowchart with Timeout

8.3.3 Continuous mode

The state flow chart of the continuous receive mode is shown in Figure 8-5.

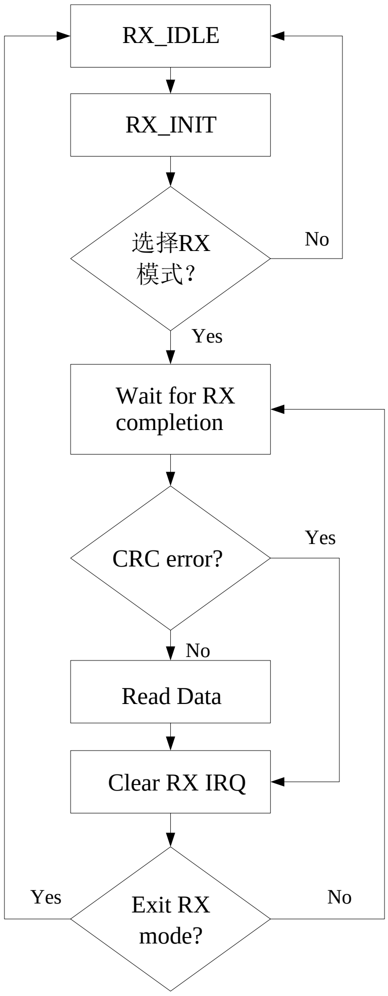


Figure 8-5 Continuous Reception Mode Flowchart

The difference between the continuous receiving mode and the single packet receiving mode and the single packet receiving mode with timeout is that after the data is received, an IRQ signal is sent to the MCU. After the MCU clears the IRQ, the next data reception begins. To exit the continuous receiving mode, the mode selection register selects the crystal oscillator output mode, and then turns off the LDO module, PLL module and other analog module circuits in turn, ending a continuous receiving mode process.

8.4 IRQ Interrupt

All interrupts of the chip share one external interrupt IO pin IRQ, and the interrupt pin is valid at high level. The chip has 7 types of interrupts, as follows. Interrupts can be enabled through the interrupt enable register, and the interrupt status can be read through the interrupt status register.

Interrupt Enable Register	Description
REG_MAPM_IRQ_MASK	IRQ MASK enable in MAPM mode 0: Generate IRQ interrupt of MAPM 1: Do not generate IRQ interrupt of MAPM
REG_RX_HMP_ERR_MASK	ECC error IRQ interrupt mask 0: Do not generate ECC error IRQ interrupt 1: Generate ECC error IRQ interrupt
REG_RX_PLHD_DONE_MASK	Early interrupt mask 0: Do not generate an IRQ interrupt in early interrupt mode 1: Generate an IRQ interrupt in early interrupt mode
REG_RX_DONE_MASK	RX complete IRQ mask 0: Do not generate RX complete IRQ interrupt 1: Generate RX complete IRQ interrupt
REG_RX_PL_CRC_ERR_MASK	RX payload CRC error IRQ mask 0: Do not generate Rx payload CRC error IRQ interrupt 1: Generate RX payload CRC error IRQ interrupt
REG_RX_TIMEOUT_MASK	RX timeout IRQ mask 0: Do not generate RX timeout IRQ interrupt 1: Generate RX timeout IRQ interrupt
REG_TX_DONE_MASK	TX completion irq mask 0: Do not generate TX completion IRQ interrupt 1: Generate TX completion IRQ interrupt

Interrupt Status Register	Description
REG_MDM_MAPM_IRQ	MAPM IRQ interrupt
REG_RX_HM_P_ERR_IRQ	ECC error IRQ interrupt
REG_RX_PLHD_DONE_IRQ	Early interrupt mode IRQ interrupt
REG_RX_DONE_IRQ	RX Completed IRQ Interrupt
REG_RX_PL_CRC_ERR_IRQ	RX payload CRC error IRQ interrupt
REG_RX_TIMEOUT_IRQ	RX timeout IRQ interrupt
REG_TX_DONE_IRQ	TX Completed IRQ Interrupt

8.5 CAD-IRQ Interrupt

The chip CAD-IRQ interrupt pin reuses the GPIO11 pin. When using it, you need to configure the GPIO11 hardware to CAD-IRQ output mode.

Register Name	Register Location	Description
REG_OUT_FUNC_SEL_CAD	page0，0x5e，bit6	CAD function selection When REG_OUT_FUNC_SEL_CAD is 1, GPIO is a normal IO, and its output can be configured by register PAD_OUT_REG; when REG_OUT_FUNC_SEL_CAD is 0, GPIO output is CAD function pin.
PAD_OUT_REG_CAD	page0，0x68，bit3	Register that controls GPIO11 output.

9 Interface design

The chip can read and write registers and transceiver FIFOs through four-wire SPI, three-wire SPI and I2C. By default, it supports both four-wire SPI and I2C. The three-wire SPI needs to be configured through the register SPI3_EN_REG to take effect.

Note: FIFO address is a special area. The register address of this area is the entry address of these special areas. The operation method is different from that of regular registers. For the FIFO operation method, please refer to Chapter 9.4 FIFO.

9.1 Four-wire SPI

The chip implements the SPI bus slave, which is used to read and write registers and FIFO. The SPI bus is a four-wire system, which are:

- SCK (clock)
- CSN (Chip select signal, low level active)
- MOSI (Data input)
- MISO (Data output)

Among them, SCK, CSN, and MOSI are controlled by the host Master, and MISO is controlled by the Slave.

During the communication process, the CSN level is pulled low to start, and the transmission process ends when the CSN level is pulled high. The host Master sends data through MOSI and MISO receives data. Data is generated at the falling edge of SCK and data sampling is performed at the rising edge.

The information transmitted by the Master consists of two parts: Address Byte and Data Byte. The first 7 bits of Address Byte are the address bits addr; the last 1 bit is the read/write bit wr, which is set to 1 during a write operation and set to 0 during a read operation.

SPI has three transmission modes:

- Single: Single byte transmission mode. The information is only 2 bytes, and the Master senys the Address Byte through MOSI.

If it is a write operation, the Master continues to send the Data Byte through MOSI; if it is a read operation, the Master reads the Data Byte replied by the Slave on MISO.

- Burst: Burst continuous transmission mode. When the information is larger than 2 bytes, the Address Byte is followed by several Data Bytes. There is no need to add Address Bytes between Data Bytes. The Slave will automatically increment the address between each Data Byte. The CSN signal is pulled high after the last Data Byte, and the rest of the information transmission process is maintained at a low level.

- FIFO: FIFO read/write mode. In this mode, single byte or continuous transmission can be realized. The transmission rules are the same as Single mode and Burst mode. The difference is that the address bit addr in Address Byte can only be configured as 0x01, and Slave does not perform address increment operation between Data Bytes.

In addition, by configuring the REG_OUT_PAD_MODE register, MISO can be configured to input high-impedance state to achieve a 4-wire SPI one-master-multiple-slave connection architecture.

9.1.1 Four-wire SPI write timing

The SPI write timing is as follows:

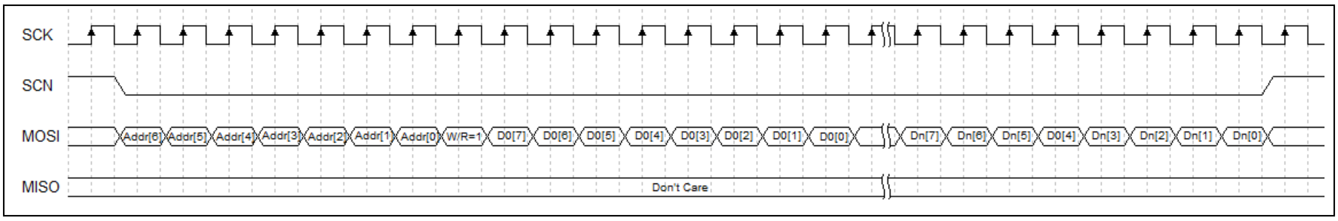


Figure 9-1 SPI Write Timing

9.1.2 Four-wire SPI read timing

The SPI read timing is as follows:

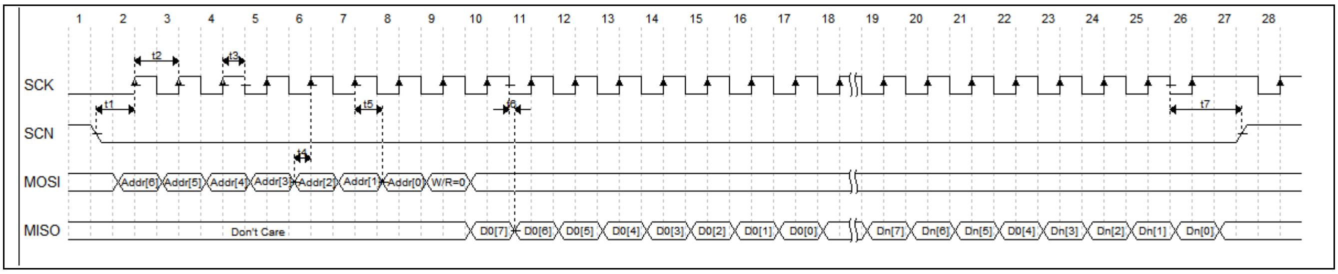


Figure 9-2 SPI Read Timing

9.1.3 Four-wire SPI timing requirements

Parameter	Description	Minimum	Typical	Maximum	Unit
t1	SCN falling edge to SCK rising edge	32	-	-	ns
t2	SCK clock period	100	-	-	ns
t3	SCK high time	50	-	-	ns
t4	Setup time from MOSI to SCK clock	7	-	-	ns
t5	Hold time from MOSI to SCK clock	7	-	-	ns
t6	SCK falling edge to MISO delay	0	-	15	ns
t7	SCK to SCN rising edge time	0	-	150	ns

9.2 Three-wire SPI

The chip supports four-wire SPI and I2C by default. If you need to use three-wire SPI, you need to configure the register SPI3_EN_REG after power-on to make the three-wire SPI effective. The three-wire SPI signals include:

- SCK (clock)
- CSN (chip select signal, low level active)
- MOSI (Data Input/Output)

9.2.1 Three-wire SPI write timing

The SPI write timing is as follows:

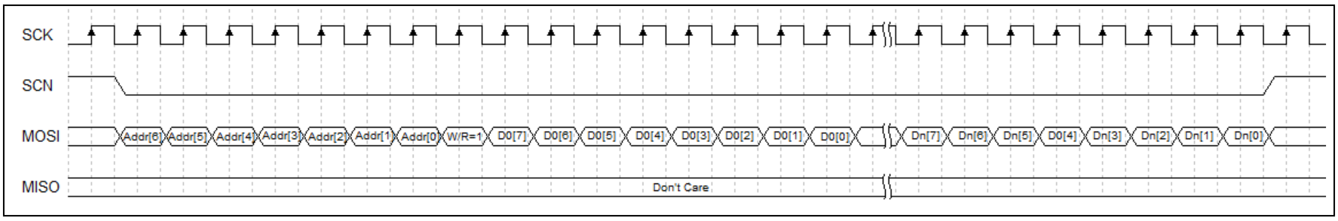


Figure 9-3 3-wire SPI write timing

9.2.2 Three-wire SPI read timing

The SPI read timing is as follows:

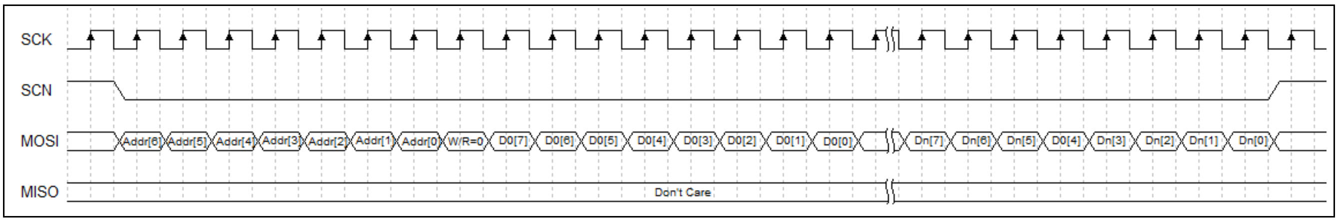


Figure 9-4 3-wire SPI read timing

9.2.3 Three-wire SPI timing requirements

Please refer to Section 9.1.3.

9.3 I2C

The I2C device address width is 7 bits and the value is 0x72.

The timing diagram of the I2C start signal Start is as follows:

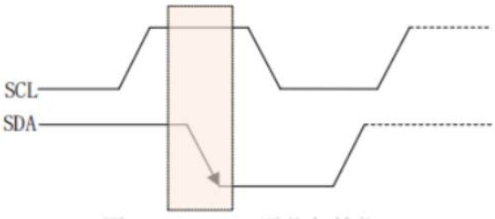


Figure 9-5 I2C start signal

The timing diagram of the I2C termination signal Stop is as follows:

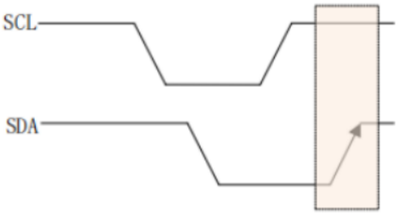


Figure 9-6 I2C stop signal

9.3.1 I2C Write Timing

The I2C write timing is as follows:

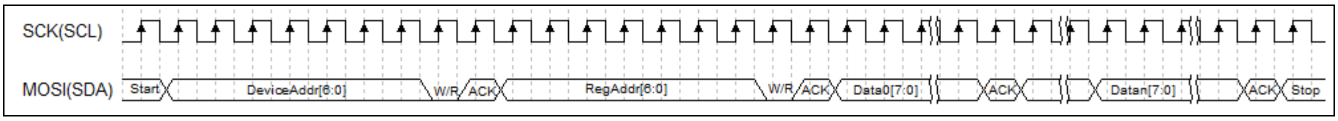


Figure 9-7 I2C write timing

9.3.2 I2C Read Timing

The I2C read timing is as follows:

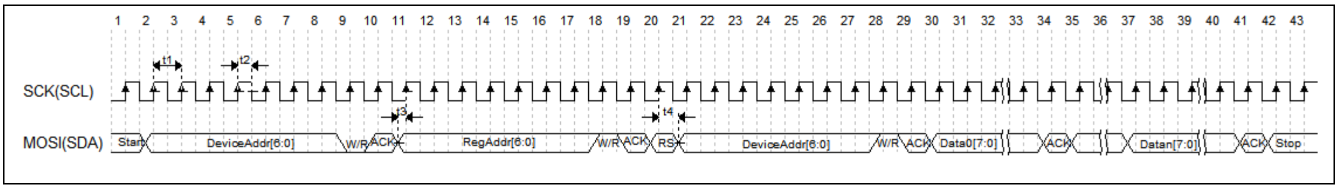


Figure 9-8 I2C read timing

9.3.3 I2C Timing Requirements

Parameter	Description	Minimum	Typical	Maximum	Unit
t1	SCL clock period	200	-	-	ns
t2	SCL high time	100	-	-	ns
t3	Setup time of SDA to SCL clock	12	-	-	ns
t4	Hold time of SDA to SCL clock	12	-	-	ns

9.4 FIFO

The chip has a 256-byte FIFO to store the TX module's transmitted data and the RX module's decoded data.

The FIFO consists of a single-port RAM and can only store and read a single packet of data information. If there is already a packet of data in the FIFO, the data should be read before writing, otherwise the previous packet of data in the FIFO will be overwritten.

In the working mode of STB3 and later, the FIFO can be read and written by Modem, SPI, and I2C.

The read and write address of the FIFO is 0x01, and the FIFO content can be read and written through this address through SPI or I2C.

9.5 GPIO

The chip supports multiple GPIO input and output controls, and the GPIO can be controlled through register configuration. For register configuration, please refer to Section 12.4. Each GPIO port can be configured as internal pull-up/pull-down input, high-impedance input (floating input), push-pull output (CMOS output) and other modes. The configuration method is shown in Table 9-1.

Table 9-1 GPIO port mode configuration

DIEN	OE	OUT	PUEN	PDEN	IO direction	IO state
0	1	0/1	0	0	Push-pull output	0/1
0	0	x	0	0	Open-drain output	High impedance
1	0	x	0	0	Input	High impedance
1	0	x	1	0	Input	Pull-up
1	0	x	0	1	Input	Pull-down

Note: The output modes of GPIO10 and GPIO11 in the chip can be reused to implement special functions.

9.5.1 External PA control GPIO

GPIO10 has two configuration output modes: normal IO and PA control pin. The different modes of GPIO10 can be configured through the register REG_OUT_FUNC_SEL_PA.

When GPIO10 is configured as a normal IO, it can output high and low levels, just like other GPIOs.

When GPIO10 is configured as PA control, and the external control PA mode is configured to the on state through the register, GPIO10 can be used as the enable control pin of the external PA, that is, GPIO10 only outputs a high level when sending data in the transmission mode, and outputs a low level at other times.

Register Name	Register Location	Description
REG_OUT_FUNC_SEL_PA	page0，0x5e，bit7	GPIO10 function selection When REG_OUT_FUNC_SEL_PA is 1, GPIO10 is a normal IO, and its output can be configured by register PAD_OUT_REG_PA; whenREG_OUT_FUNC_SEL_PA is 0, GPIO10 output is an external PA function pin.
PAD_OUT_REG_PA	page0，0x68，bit2	GPIO10 output control Register. When REG_OUT_FUNC_SEL_PA is 1, GPIO10 is a normal IO, and its output can be configured by register PAD_OUT_REG_PA.

9.5.2 CAD monitoring GPIO

GPIO11 has two configuration modes: normal IO and CAD detection pin. The different modes of GPIO11 can be configured through the register REG_OUT_FUNC_SEL_CAD.

When GPIO11 is configured as a normal IO, it can output high and low levels, just like other GPIOs.

When GPIO11 is configured as a CAD signal detection interrupt output, and the CAD function is also configured to be turned on through the register, and the chip is in the RX state, when receiving a signal with the same bandwidth and spreading factor parameters as the currently configured, GPIO11 will generate a continuous and stable high level, indicating that the chip has searched for an air signal, and the high level will not be pulled down until the signal disappears; in other situations and modes, GPIO11 outputs a low level. For specific CAD configuration and use, please refer to Chapter 9.4.

The chip CAD-IRQ interrupt pin reuses the GPIO11 pin. When using it, you need to configure the GPIO11 hardware to CAD-IRQ output mode.

Register Name	Register Location	Description
REG_OUT_FUNC_SEL_CAD	page0，0x5e，bit6	CAD function selection When REG_OUT_FUNC_SEL_CAD is 1, GPIO is a normal IO, and its output can be configured by register PAD_OUT_REG; when REG_OUT_FUNC_SEL_CAD is 0, GPIO output is CAD function pin.
PAD_OUT_REG_CAD	page0，0x68，bit3	GPIO11 output control register. When REG_OUT_FUNC_SEL_CAD is 1, GPIO11 is a normal IO, and its output can be configured by register PAD_OUT_REG_CAD.

10 ChirpIoT special function description

10.1 RSSI Function

10.1.1 ChirpIoT signal RSSI indication

The chip supports the signal energy statistics function. When the chip receives ChirpIoT data, the internal logic of the chip will count the signal strength RSSI value of the current data packet. However, after the data packet is received, the MCU can read the register RSSI_R32_HOLD or RSSI_RBW_HOLD through the SPI interface to obtain the RSSI value of the current data packet.

10.1.2 Channel detection RSSI indication

The RSSI function also supports energy statistics indication when detecting channels. After the chip enters the RX state, it can perform RSSI energy statistics once by configuring the register REG_RSSI_PRE_RD, and then read the current signal RSSI energy through the register RSSI_R32_HOLD.

REG_RSSI_PRE_RD is first configured to 0 and then to 1 to update the RSSI value once. The register location is as follows:

Register Name	Register Location	Description
REG_RSSI_PRE_RD	page2，0x06，bit2	Enable channel RSSI reading detection and update the RSSI value

10.2 Signal-to-noise energy statistics function

The chip supports signal noise statistics. When the chip receives ChirpIoT data, the internal logic of the chip will calculate the signal energy and noise energy of the current data packet. However, after the data packet is received, the MCU can read the signal energy value of the register SIG_POW_AVG and the noise energy value of the register NOI_POW_AVG through the SPI interface, and then the MCU can calculate the signal-to-noise ratio of the data packet through the standard formula.

10.3 Channel Activity Detection (CAD)

Since ChirpIoT™ signals have excellent receiving performance, they can still ensure normal signal reception and demodulation when the signal strength is lower than the receiver noise floor. In this case, the method of using only the signal strength indicator RSSI to determine whether the channel is occupied can no longer meet the needs. Therefore, the chip is specially designed with a channel activity detector to detect ChirpIoT™ signals, ensuring that the channel can be found to be occupied by ChirpIoT™ signals sent by other devices when the signal strength is lower than the receiver noise floor. The detection function of the channel interaction detector is divided into two parts, one is the ChirpIoT™ signal preamble detection, and the other is the ChirpIoT™ signal data segment detection.

10.3.1 ChirpIoT signal leading segment detection

As shown in Figure 10-1:

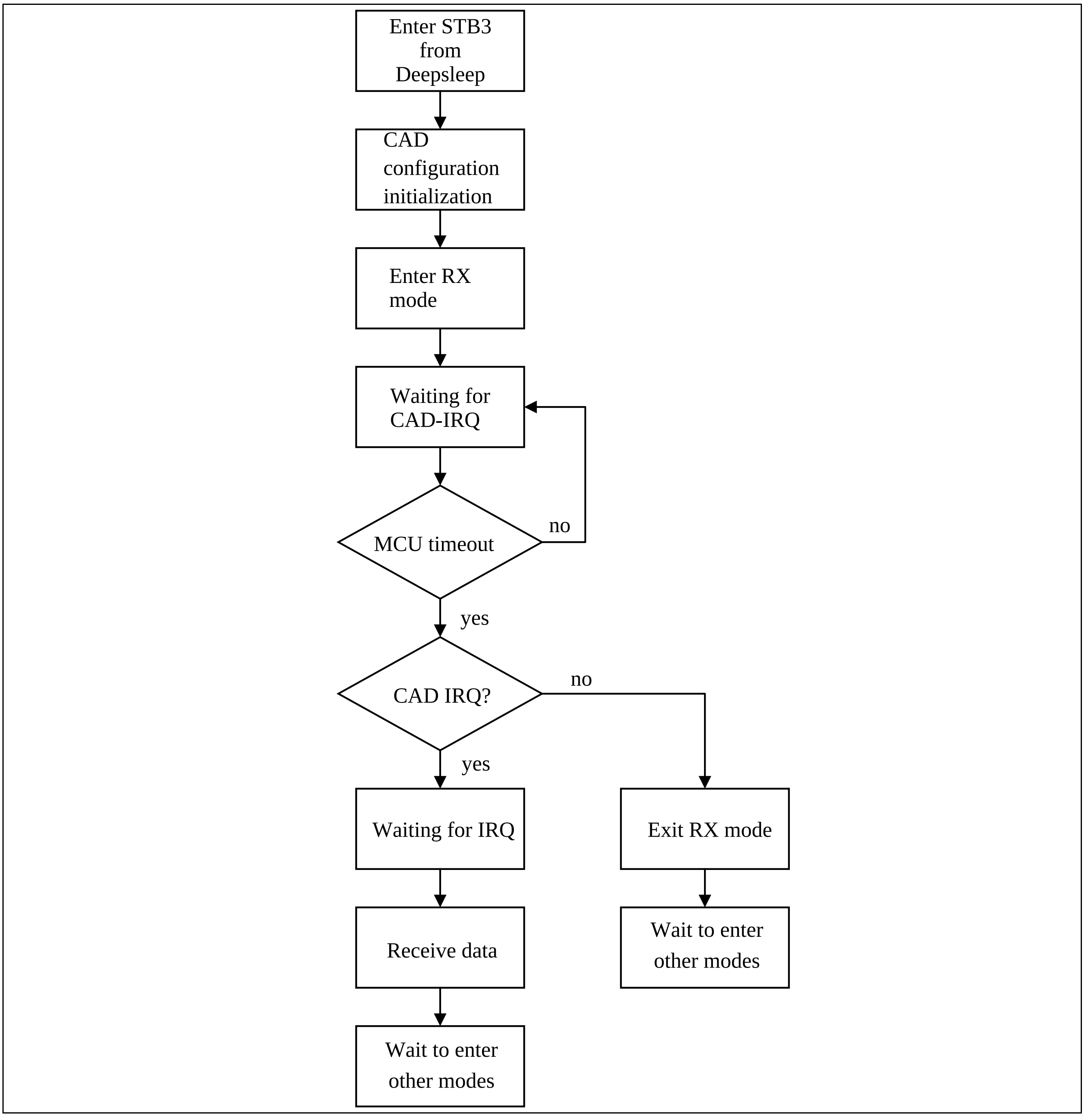


Figure 10-1 Channel Activity Detection (CAD)

The chip supports CAD-IRQ interrupt. After entering RX mode, the chip will detect whether there is a leading signal of ChirpIoT™ in the channel. If so, CAD-IRQ will be set high. The external MCU can determine whether there is a leading signal of ChirpIoT™ in the channel by detecting whether the CAD-IRQ signal is pulled high within a certain period of time. If so, it can continue to wait for the IRQ interrupt of data reception and then receive data; if there is no leading signal of ChirpIoT™, Then it quickly exits RX mode and waits for commands to enter other modes.

In addition, in order to reduce false triggering of CAD-IRQ, a filtering function of CAD-IRQ is added, which can be enabled by configuring the register.

Address	Register	Bit	Default	Access	Description
[1][0x25]	REG_DEC_VAL_THRE	1:0	0x3	W/R	CAD 检测阈值，规定连续接收到几个 preamble 之后 CAD 拉高。

10.3.2 ChirpIoT signal data segment detection

In order to avoid missing the leading signal of the ChirpIoT signal and thus failing to achieve effective CAD detection during the actual reception process, the chip also supports the detection of non-ChirpIoT leading signals, namely the ChirpIoT data segment signal detection function.

The data segment signal detection can be started by configuring the registers REG_SIGNAL_POWER_THRED, REG_DEC_PLD_THRE, and REG_DEC_PLD_EN.

Register configuration description:

Address	Register	Bit	Default	Access	Description
[1][0x35]	REG_SIGNAL_POWER_THRED	7:4	4'b1110	W/R	CAD 有效信号的阈值，只有能量大于此值才能进入 CAD 有效信号的判断。
	REG_DEC_PLD_THRE	3:2	2'b10	W/R	CAD 有效信号个数阈值，只有接收到满足条件的此数值个数的 preamble，才能拉高 CAD。
	REG_DEC_PLD_EN	1	1'b0	W/R	CAD payload 检测使能 0: disable 1:enable

Note: CAD detection of the leading segment and the data segment can be turned on at the same time to ensure that data at different stages can be detected and checked.

10.4 MAPM function

In the ChirpIoT star networking scenario, in order to ensure that the node-end device can receive the data sent by the gateway, the gateway will use an extra-long preamble for transmission, such as a preamble that takes 1 second. In this case, all node-end devices will continue to receive after receiving the preamble until the node network address is displayed in the payload stage. At this time, the software of the node-end device can compare the received network address with the node device's own address. If they are different, it will stop receiving to reduce power consumption. If the address reception conditions are met, it will continue receiving. There are several problems with this solution:

① Node devices that do not meet the address receiving conditions can only stop receiving until the payload stage, wasting a lot of power.

② Node devices that meet the address receiving conditions also need to receive the extra-long preamble before entering the payload receiving phase, which also wastes power.

To solve the above two power consumption problems, the chip adds a new frame structure mode Multiple Address Preamble Mode (MAPM). This frame structure mode inserts a special address in the Preamble stage, so that the node end can judge in advance whether the address reception conditions are met based on the address in the Preamble stage, which solves problem ①.

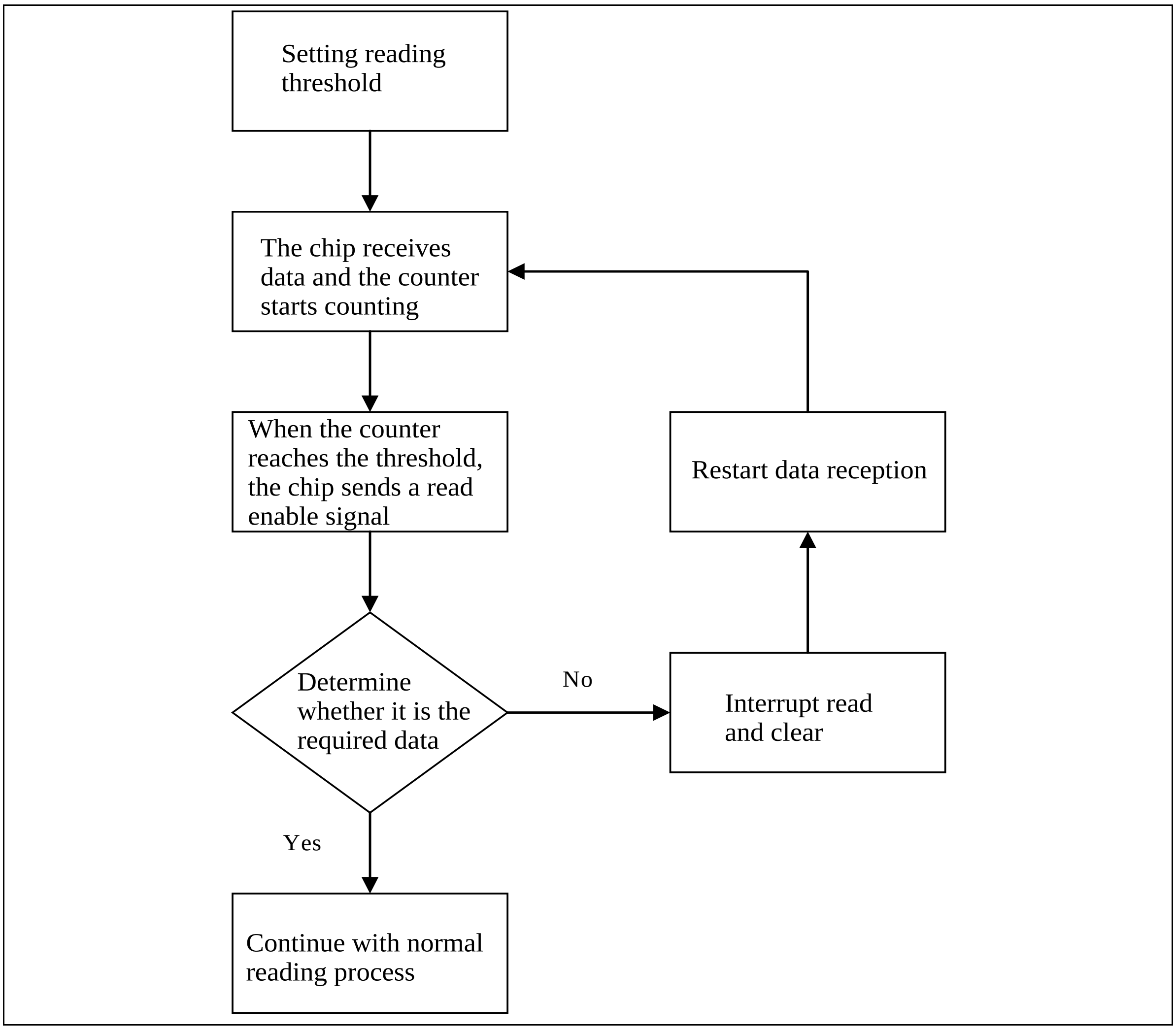
In addition, in order to solve the second problem, a variable counter is added to the special address to indicate how much preamble length is left. The node-end device that meets the address reception conditions can enter the low-power sleep state in advance. Then, the counter is used to determine how long in advance to enter the receiving state to ensure normal data reception, so as to further save the power consumption of the node-end device that meets the address reception conditions.

10.5 eFuse Function

The chip supports 82Bytes of eFuse space for customers to use, which can be used to store data that will not be lost when power is off.

10.6 Early interrupt function

The chip early interrupt function is to check the decoded data during the process of the chip reading a frame of data, determine whether it is what you want, and then decide whether to continue reading or abandon this frame of data.
The flow chart is as follows:



10.7 Intelligent SF recognition

In order to realize lightweight gateway equipment, the chip is designed with intelligent SF recognition hardware circuit, which can complete intelligent recognition according to the SF mode of ChirpIoT™ in the actual channel at the same frequency point, and automatically configure the chip SF parameters to achieve the purpose of receiving different SF signal data.

The specific implementation process is shown in Figure 10-2:

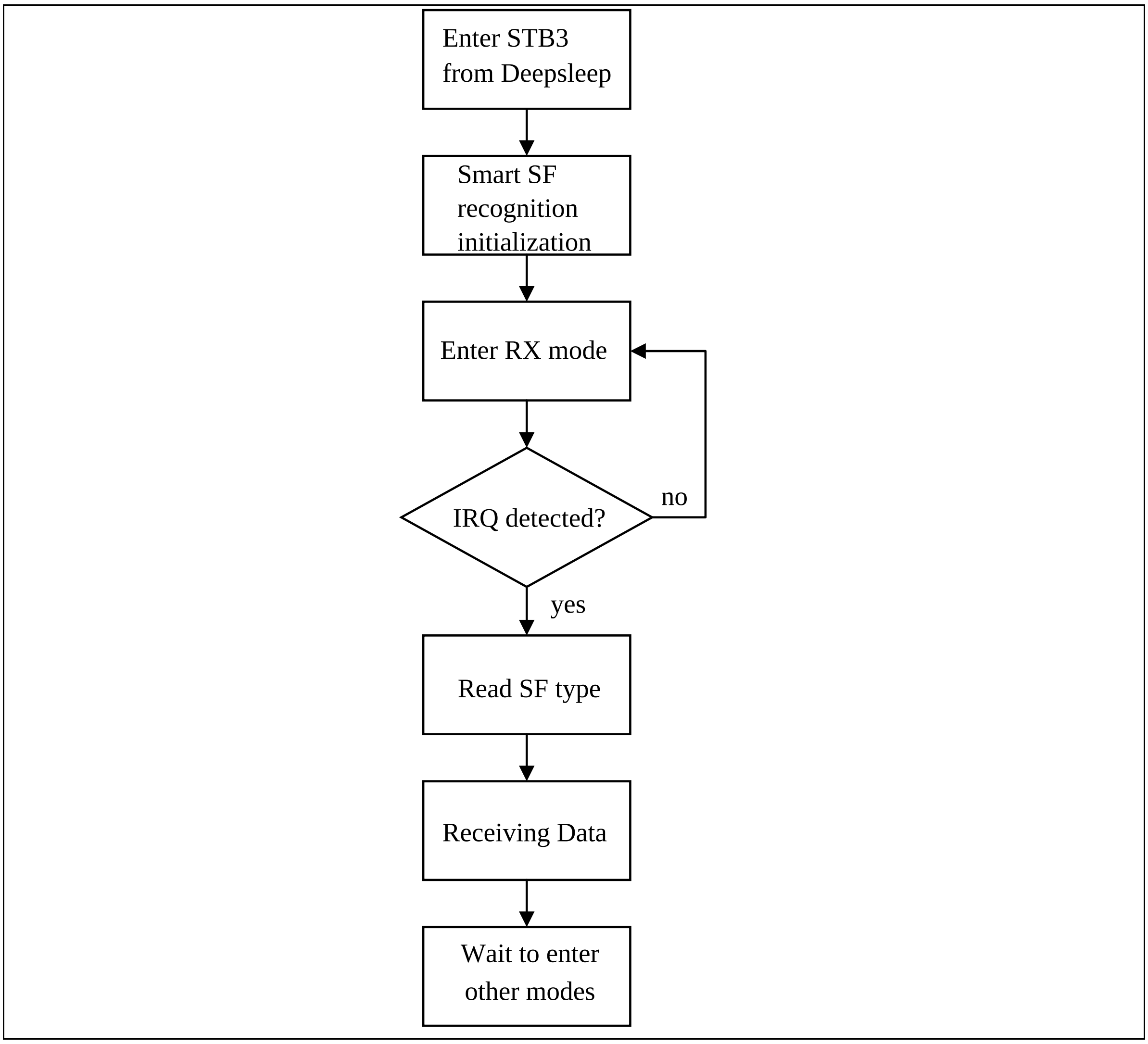


Figure 10-2 Intelligent SF identification

11 Operation Mode Description

11.1 Overview

The chip has 7 operating modes in total: DeepSleep mode, Sleep mode, STB1 mode, STB2 mode, STB3 mode, TX mode and RX mode.

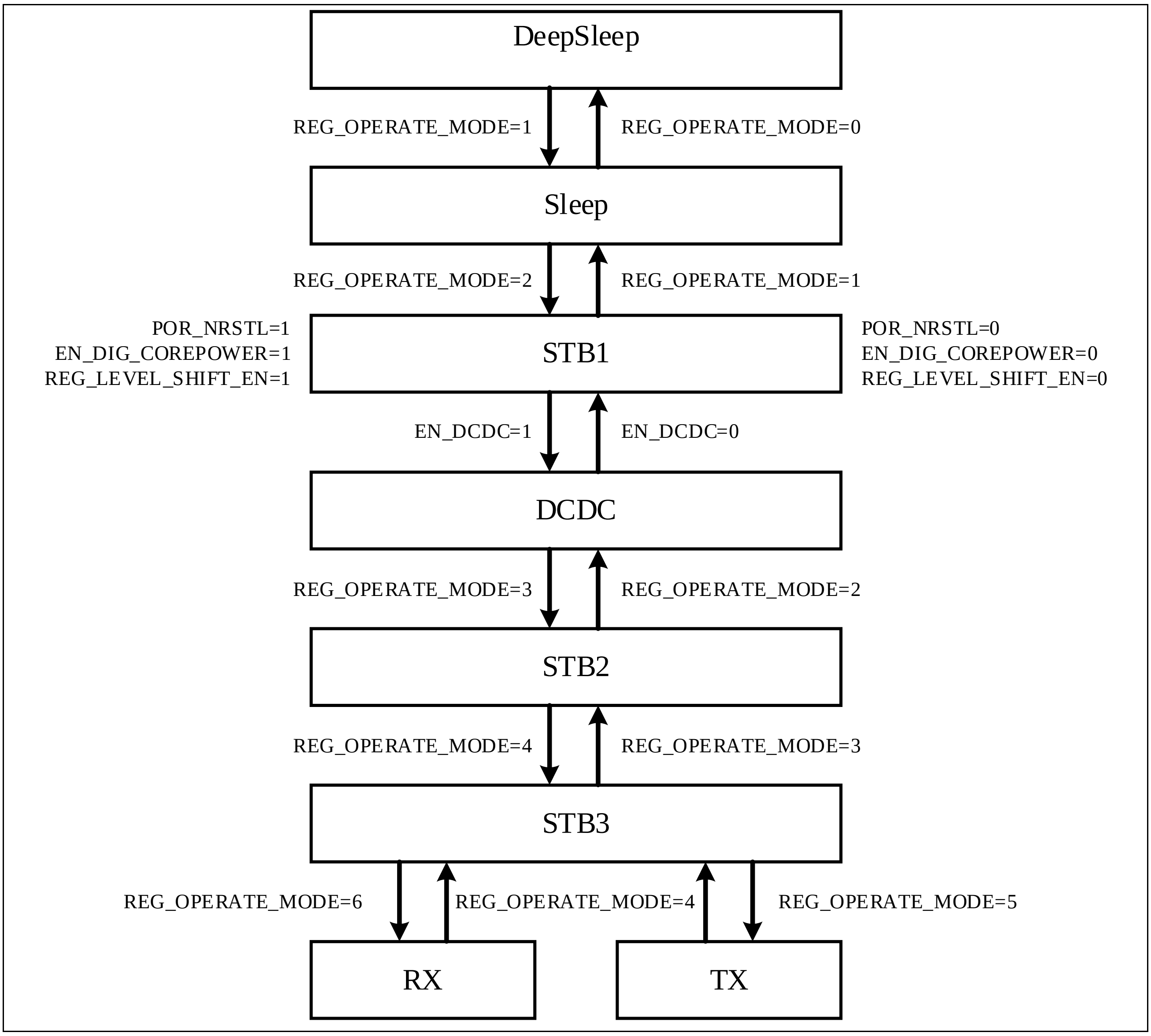


Figure 11-1 Working mode setting process

After the chip is powered on, there are seven states, including Deep Sleep mode (DeepSleep), Sleep mode (Sleep), LDO working mode (STB1), OSC working mode (STB2), OSC output mode (STB3), TX mode and RX mode. All states can be switched through the configuration register.

Note: If the chip peripheral circuit does not support DCDC, you do not need to configure EN_DCDC and directly enter STB2 from STB1. For chip circuits, refer to Chapter 13.

11.2 Working modes description

11.2.1 Deep Sleep mode

In this mode, the 3V logic area works. Except for the 3V logic area registers that can be configured, other modules are inoperable to keep the chip power consumption to a minimum.

11.2.2 Sleep mode

In this mode, low power LDO is enabled, the chip power consumption is low, and all registers can be supported to maintain power. However, configuration of registers other than the 3V logic area through SPI/I2C is prohibited.

11.2.3 STB1 mode

In this mode, the high-performance LDO starts working and the low-voltage area logic is enabled.

11.2.4 DCDC mode

In STB1 state, the DCDC function of the chip can be turned on by writing a high level to the EN_DCDC register.

Note: DCDC mode requires related peripheral circuit support, please refer to Chapter 13.

11.2.5 STB2 mode

This mode turns on the OSC crystal oscillator. The OSC starts to work and maintains oscillation but does not output to other modules. The power consumption is relatively small.

11.2.6 STB3 mode

In this mode, the OSC clock is output to each module and the chip starts to work normally.

11.2.7 TX Mode

This mode has enabled all TX transmission-related modules, enabling the encoding and sending of packet data.

11.2.8 RX Mode

This mode turns on all RX reception-related modules, waiting for the reception and decoding of packet data.

12 Registers

The chip registers are divided into 4 pages, each page has 128 register addresses. The register page is selected through the REG_PAGE_SEL of register 0x00. The registers at addresses 0x00~0x05 can be operated in any register page. The following table describes the location and function of each register of the chip.

Note:

Reading and writing of the registers in the Reserved section are prohibited.

All registers in the low-voltage area only support operation in STB1, STB2, STB3, Tx and Rx modes.

12.1 System Configuration

Page	Addr.	Register	Bit	Default	Access	Description
0/1/2/3	0x00	REG_SOFT_RST	7	0x00	W/R	Soft reset, except 1.2 V configuration registers 1.2 V Logic reset 0: Reset 1: No reset
		Reserved	6:2		W/R	Reserved
		REG_PAGE_SEL	1:0		W/R	0: Page0 1: Page1 2: Page2 3: Page3
	0x01	FIFO READ/ WRITE ACESS POINT	7:0	-	W/R	FIFO Read and write address

12.2 Mode Configuration

Page	Addr.	Register	Bit	Default	Access	Description
0/1/2/3	0x02	Reserved	7:3	0x00	W/R	Reserved
		REG_OPERATE_MODE	2:0		W/R	0: 深度睡眠模式 1: 睡眠模式 2: STB1 模式(LDO 工作) 3: STB2 模式(OSC 工作) 4: STB3 模式(OSC 输出) 5: Tx 模式 6: Rx 模式
	0x04	Reserved	7:6	0x06	W/R	Reserved
		EN_LS_3V	5		W/R	1: 电平转换使能
		POR_RSTL	4		W/R	1.2V 所有配置寄存器复位 0: 复位 1: 不复位
		Reserved	3:0		W/R	Reserved

12.3 MAC configuration (low voltage area registers)

Page	Addr.	Register	Bit	Default	Access	Description
0	0x54	Reserved	7:3	0x00	W/R	Reserved
		REG_TX_FIFO_FULL	2		W/R	0: TX FIFO 未满 1: TX FIFO 已满
		REG_TX_FIFO_FULL_SEL	1		W/R	0: select fifo wr done 1: select reg tx fifo full
		Reserved	0		W/R	Reserved
	0x58	REG_RX_TIMEOUT_IMMED	7	0x0F	W/R	1: Rx 超时 IRQ 立即生成
		REG_MAPM_IRQ_MASK	6		W/R	MAPM 模式下的 IRQ MASK 使能 0: 不产生 MAPM 的 IRQ 中断 1: 产生 MAPM 的 IRQ 中断
		Reserved	5		W/R	Reserved
		REG_RX_PLHD_DONE_MASK	4		W/R	提前中断模式掩码 0: 不产生提前中断模式的 IRQ 中断 1: 产生提前中断模式的 IRQ 中断
		REG_RX_DONE_MASK	3		W/R	rx 完成 irq 掩码 0: 不产生 rx 完成的 IRQ 中断 1: 产生 rx 完成的 IRQ 中断
		REG_RX_PL_CRC_ERR_MASK	2		W/R	Rx 有效负载 crc 错误 irq 掩码 0: 不产生 Rx 有效负载 crc 错误的 IRQ 中断 1: 产生 Rx 有效负载 crc 错误的 IRQ 中断
		REG_RX_TIMEOUT_MASK	1		W/R	Rx 超时 irq 掩码 0: 不产生 Rx 超时的 IRQ 中断 1: 产生 Rx 超时的 IRQ 中断
		REG_TX_DONE_MASK	0		W/R	Tx 完成 irq 掩码 0: 不产生 Tx 完成的 IRQ 中断 1: 产生 Tx 完成的 IRQ 中断
	0x5b	Reserved	7:3	0x03	W/R	Reserved
		REG_RX_DONE_RST_TEST	2		W/R	1: 一旦 RX 完成，modem 即刻复位 0: IRQ 清零且 RX 完成后 modem 才复位
		REG_RX_CR_ERR_RST	1		W/R	1: rx 码率错误复位使能
		REG_RX_HEADER_ER_RST	0		W/R	1: rx 包头错误复位使能
	0x5d	Reserved	7:1	0x00	W/R	Reserved
		REG_2POINT_CAL_EN	0		W/R	两点 cal 使能，高电平有效
	0x6c	EFUSE_WR_IRQ	7	0x00	R	eFuse 读写完中断 1: 高电平说明读写完成
		REG_MDM_MAPM_IRQ	6		R	MAPM 中断状态寄存器 1: MAPM IRQ；0: 非 MAPM IRQ
		Reserved	5		W/R	Reserved
		REG_RX_PLHD_DONE_IRQ	4		W/R	写 1 清零提前中断
		REG_RX_DONE_IRQ	3		W/R	写 1 清零 rx done irq
		REG_RX_PL_CRC_ERR_IRQ	2		W/R	写 1 清零 rx payload crc error irq

		REG_RX_TIMEOUT_IRQ	1		W/R	写 1 清零 rx timeout irq
		REG_TX_DONE_IRQ	0		W/R	写 1 清零 tx done irq
	0x6D	REG_MAPM_PN_CNT	7:0	0x00	R	MAPM 模式的 preamble 数量计数器
	0x6E	REG_MAPM_ADDRX	7:0	0x00	R	MAPM 模式的 addrx

12.4 GPIO interface configuration (low voltage area registers)

Page	Addr.	Register	Bit	Default	Access	Description
0	0x5e	REG_OUT_FUNC_SEL_PA	7	0x00	W/R	1: 作为普通 IO 使用 0: 作为外置 PA 使用
		REG_OUT_FUNC_SEL_CAD	6		W/R	1: 作为普通 IO 使用 0: 作为 CAD 引脚使用
		Reserved	5:0		W/R	Reserved
	0x5f	GPIO_PDEN LSB	7:0	0x00	W/R	GPIO_PDEN LSB[3]: GPIO3 下拉使能 GPIO_PDEN LSB[0]: GPIO0 下拉使能 其他配置 Reserved
	0x60	Reserved	7:4	0x00	W/R	Reserved
		GPIO_PDEN MSB	3:0		W/R	GPIO_PDEN MSB[3]: GPIO11 下拉使能 GPIO_PDEN MSB[2]: GPIO10 下拉使能 其他配置 Reserved
	0x61	GPIO_PUEN LSB	7:0	0x00	W/R	GPIO_PDEN LSB[3]: GPIO3 上拉使能 GPIO_PDEN LSB[0]: GPIO0 上拉使能 其他配置 Reserved
	0x62	Reserved	7:4	0x00	W/R	Reserved
		GPIO_PUEN MSB	3:0		W/R	GPIO_PDEN MSB[3]: GPIO11 上拉使能 GPIO_PDEN MSB[2]: GPIO10 上拉使能 其他配置 Reserved
	0x63	GPIO_DIEN LSB	7:0	0x00	W/R	GPIO_PDEN LSB[3]: GPIO3 输入使能 GPIO_PDEN LSB[0]: GPIO0 输入使能 其他配置 Reserved
	0x64	Reserved	7:4	0x00	W/R	Reserved
		GPIO_DIEN MSB	3:0		W/R	GPIO_PDEN MSB[3]: GPIO11 输入使能 GPIO_PDEN MSB[2]: GPIO10 输入使能 其他配置 Reserved
	0x65	GPIO_OE LSB	7:0	0x00	W/R	GPIO_PDEN LSB[3]: GPIO3 输出使能 GPIO_PDEN LSB[0]: GPIO0 输出使能 其他配置 Reserved
	0x66	Reserved	7:4	0x00	W/R	Reserved
		GPIO_OE MSB	3:0		W/R	GPIO_PDEN MSB[3]: GPIO11 输出使能 GPIO_PDEN MSB[2]: GPIO10 输出使能 其他配置 Reserved
	0x67	GPIO_OUT LSB	7:0	0x00	W/R	GPIO_PDEN LSB[3]: GPIO3 输出 GPIO_PDEN LSB[0]: GPIO0 输出 其他配置 Reserved
	0x68	Reserved	7:4	0x00	W/R	Reserved
		GPIO_OUT MSB	3:0		W/R	GPIO_PDEN MSB[3]: GPIO11 输出

						GPIO_PDEN MSB[2]: GPIO10 输出 其他配置 Reserved
	0x74	Reserved	7:6	-	R	Reserved
		GPIO_IN LSB	5:0	-	R	GPIO_PDEN LSB[3]: GPIO3 输入 GPIO_PDEN LSB[0]: GPIO0 输入 其他配置 Reserved
	0x75	Reserved	7:6	-	R	Reserved
		GPIO_IN MSB	5:0	-	R	GPIO_PDEN LSB[3]: GPIO11 输入 GPIO_PDEN LSB[2]: GPIO10 输入 其他配置 Reserved

12.5 Basic operation configuration

Page	Addr.	Register	Bit	Default	Access	Description
3	0x06	Reserved	7:3	0x00	W/R	Reserved
		REG_TX_CFG_MODE	2		W/R	0: 单次发送模式 1: 连续发送模式
		REG_RX_CFG_MODE	1:0		W/R	0: 单次接收模式 1: 带超时的单次接收模式 2: 连续接收模式
	0x07	REG_RX_TIMEOUT[7:0]	7:0	0x00	W/R	接收超时设置，低 8 位，单位微秒
	0x08	REG_RX_TIMEOUT[15:8]	7:0	0x00	W/R	接收超时设置，高 8 位，单位微秒
	0x0D	BW	7:4	0x98	W/R	带宽： 6: 62.5 kHz 7: 125 kHz 8: 250 kHz 9: 500 kHz
		CODING RATE	3:1		W/R	码率： 1: 4/5 2: 4/6 3: 4/7 4: 4/8
		Reserved	0		W/R	Reserved
	0x0E	SPREADING FACTOR	7:4	0x78	W/R	SF： 5: 32 chips / symbol 6: 64 chips / symbol 7: 128 chips / symbol 8: 256 chips / symbol 9: 512 chips / symbol 10: 1024 chips / symbol 11: 2048 chips / symbol 12: 4096 chips / symbol
		RXPAYLOADCRCON	3		W/R	0: CRC 关闭 1: CRC 打开
		Reserved	2:0		W/R	Reserved
	0x0F	SYNCWORD[7:4]	7:5	0x12	W/R	同步字高 4bit

						SF5 时取值范围 0x1~0x 3 SF6 时取值范围 0x 1~0x 7 其余 SF 取值范围 0x 1-0x F
		SYNCWORD[3:0]	3:0		W/R	同步字低 4bit SF5 时取值范围 0x0~0x 3 SF6 时取值范围 0x 0~0x 7 其余 SF 取值范围 0x 0-0x F
	0x10	IF[7:0]	7:0	0xC0	W/R	中频配置，次高 8 位
	0x11	IF[11:8]	7:4	0x0F	W/R	中频配置，高 4 位
		Reserved	3:0		W/R	Reserved
	0x12	Reserved	7:4	0x06	W/R	Reserved
		LOWDATARATE	3		W/R	0: 正常模式 1: 低速率模式
		Reserved	2:0		W/R	Reserved
	0x13	PREAMBLE_LEN[7:0]	7:0	0x08	W/R	preamble 长度低 8 位
	0x14	PREAMBLE_LEN[15:8]	7:0	0x00	W/R	preamble 长度高 8 位
	0x1A	SPI3_EN_REG	7	0x03	W/R	3 线 SPI 选择寄存器 0: 不选择 3 线 SPI 模式; 1: 选择 3 线 SPI 模式
		REG_MOSI_PUEN	6		W/R	mosi 管脚的寄存器控制模式 puen 控制寄存器
		REG_CSN_PUEN	5		W/R	csn 管脚的寄存器控制模式 csn 控制寄存器
		REG_SCK_PUEN	4		W/R	sck 管脚的寄存器控制模式 sck 控制寄存器
		REG_IN_PAD_MODE	3		W/R	SPI 输入管脚手动配置模式使能
		REG_OUT_PAD_MODE	2:0		W/R	SPI 的 MISO 管脚模式选择，用于 1 对多的 SPI 应用
	0x24	Reserved	7:4	0x00	W/R	Reserved
		EN_DCDC_3V	3		W/R	DCDC 使能 0: DCDC bypass 1: DCDC 工作
		Reserved	2:0		W/R	Reserved
	0x26	XTAL_ACTIVE_EN_3V	7	0x00	W/R	支持 3V 供电的有源晶振通道使能，保证芯片可工作在晶振 3V 时钟下 1: 打开；0: 关闭
		XTAL_STARTUP_EN_3V	6		W/R	晶振快速启动使能。1: 打开；0: 关闭。
		EN_DIG_COREPOWER_3V	5		W/R	数字核电压使能开关 1: 打开；0: 关闭
		Reserved	4:0		W/R	Reserved

13 Reference Schematic

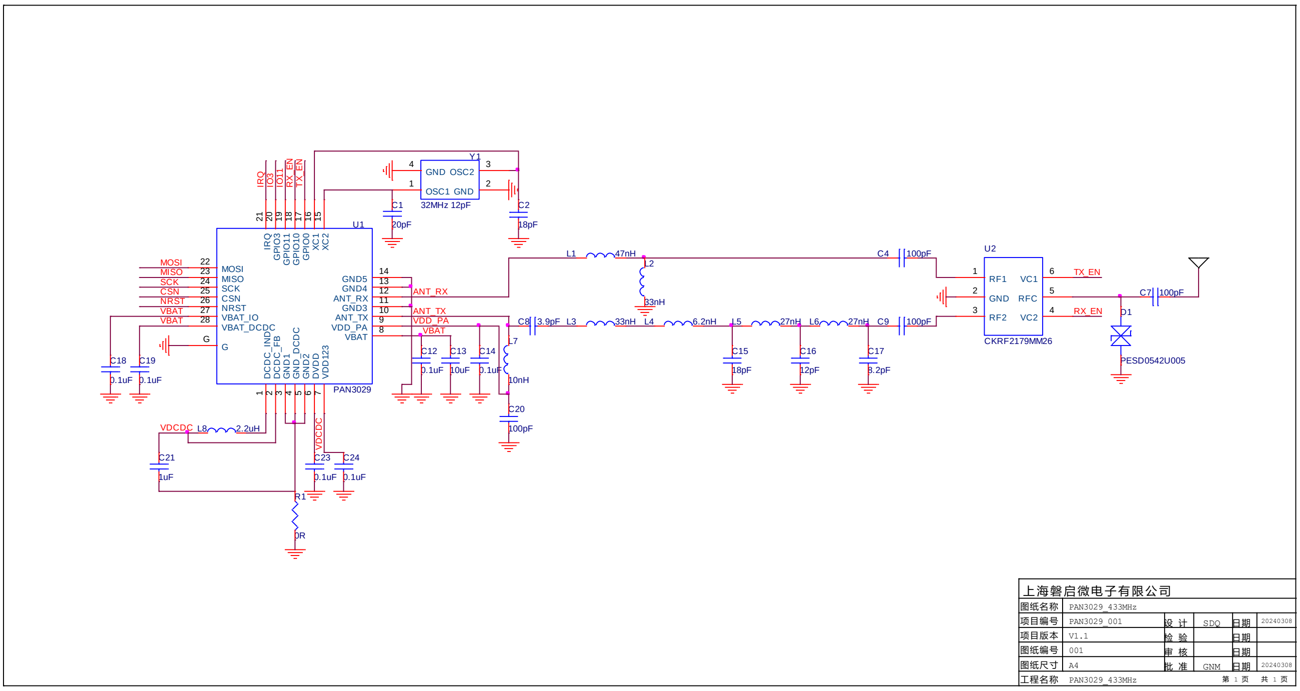


Figure 13-1 433 MHz reference schematic

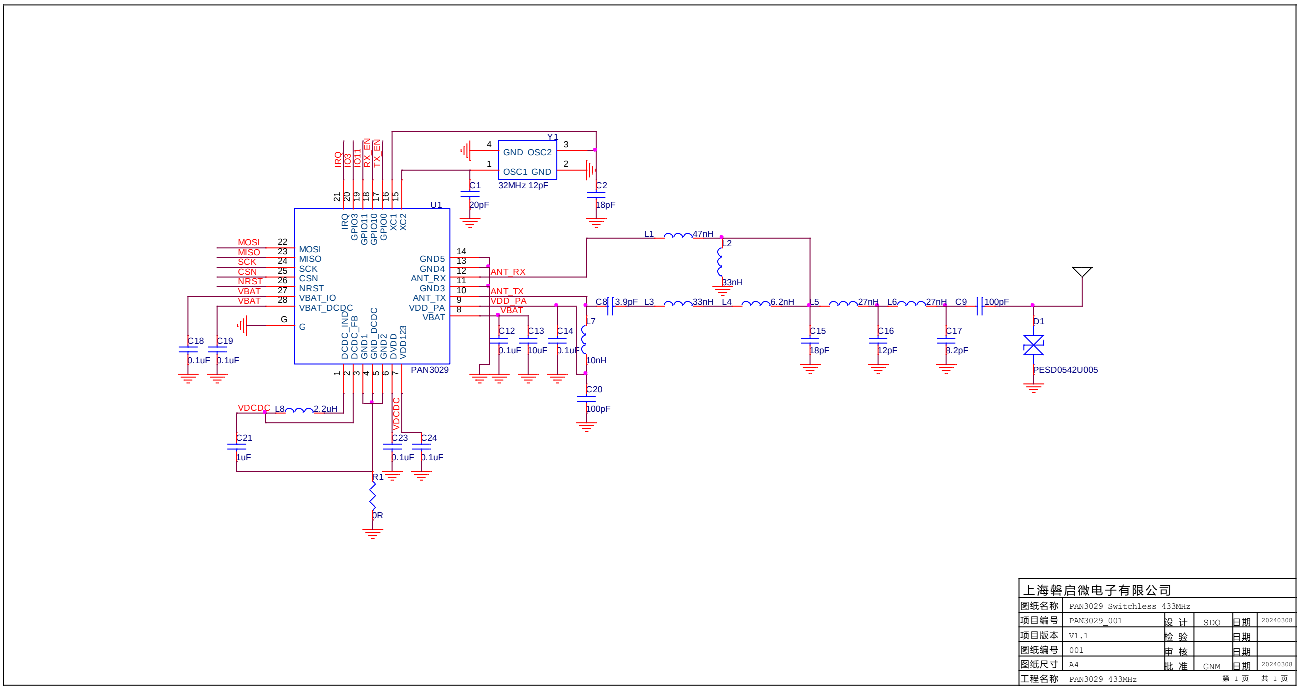


Figure 13-2 Switchless 433MHz reference schematic

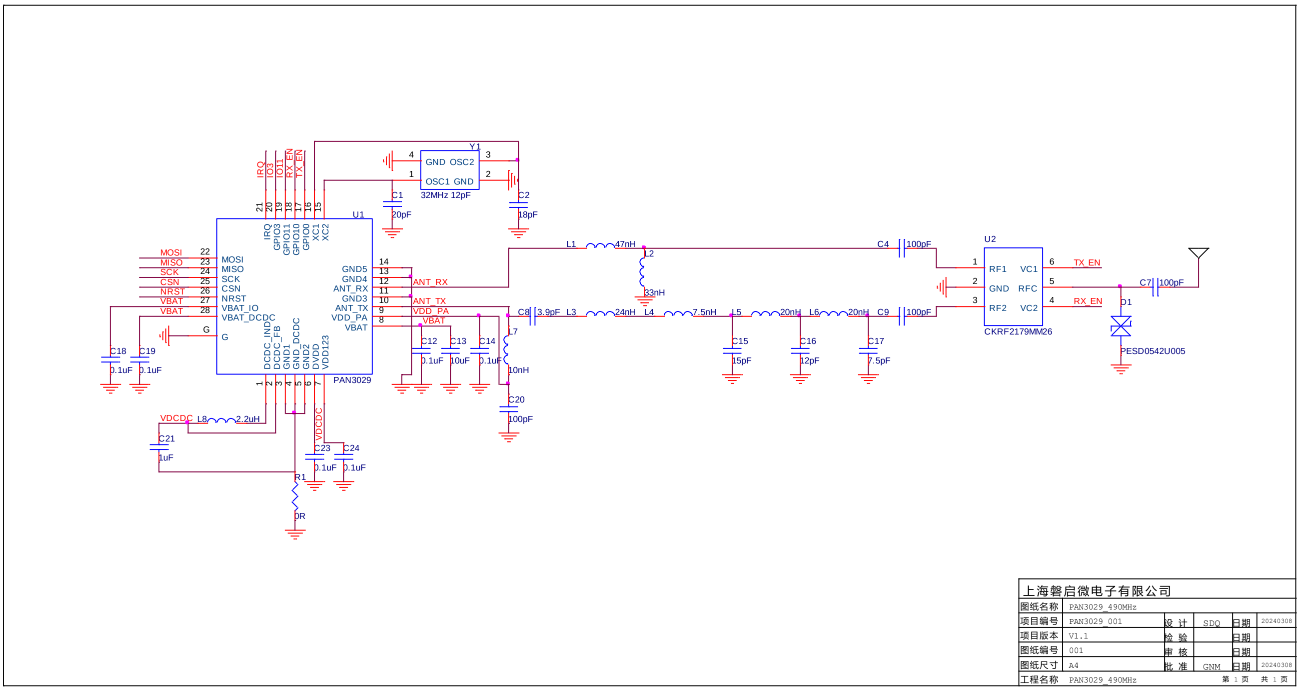


Figure 13-3 490 MHz reference schematic

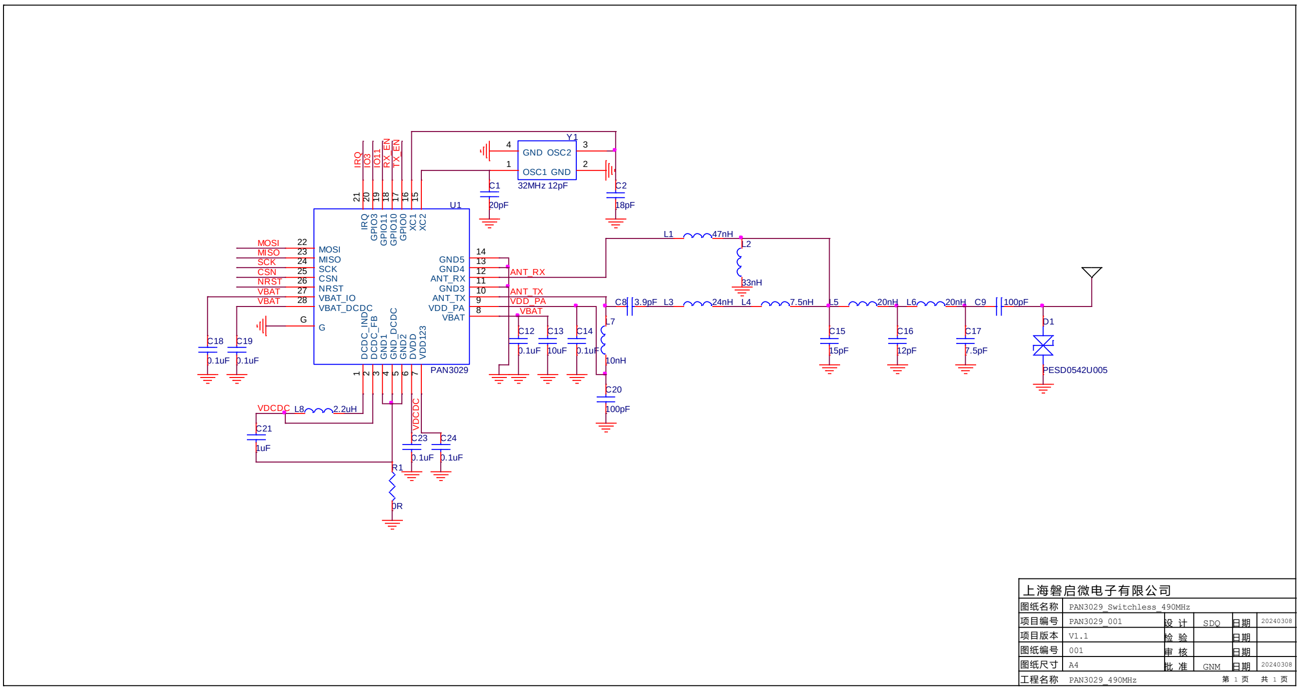


Figure 13-4 Switchless 490 MHz reference schematic

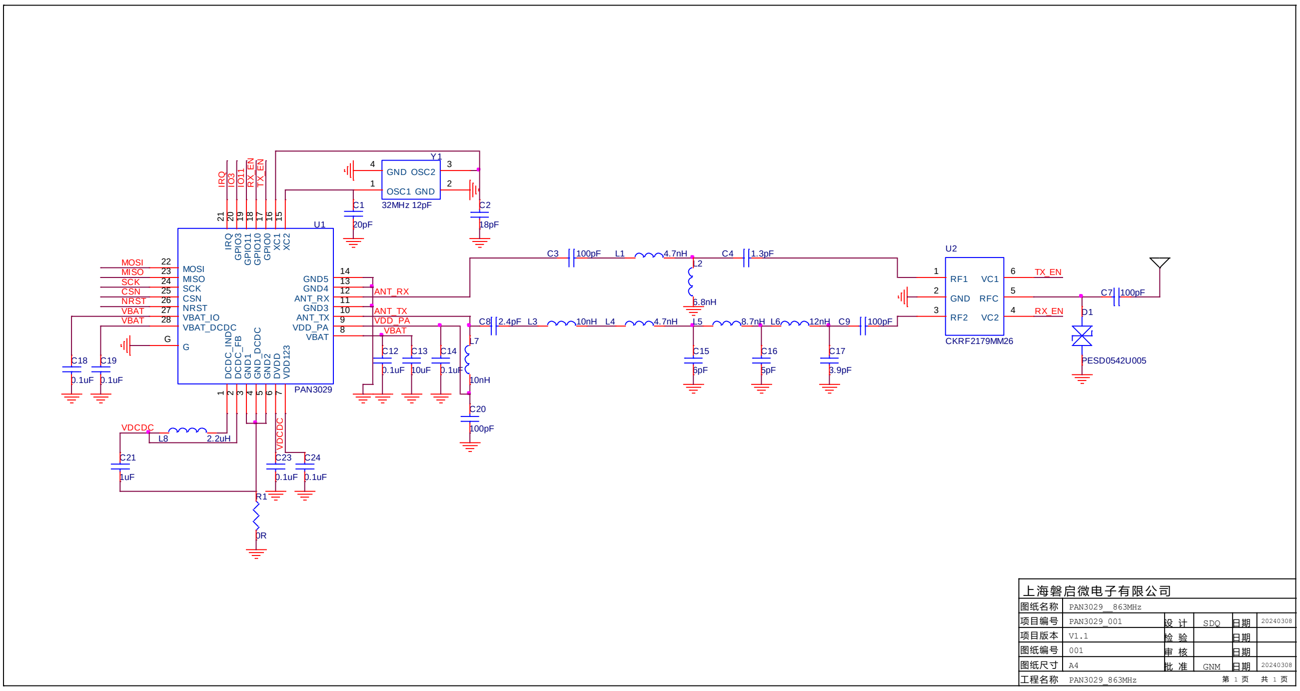


Figure 13-5 863 MHz reference schematic

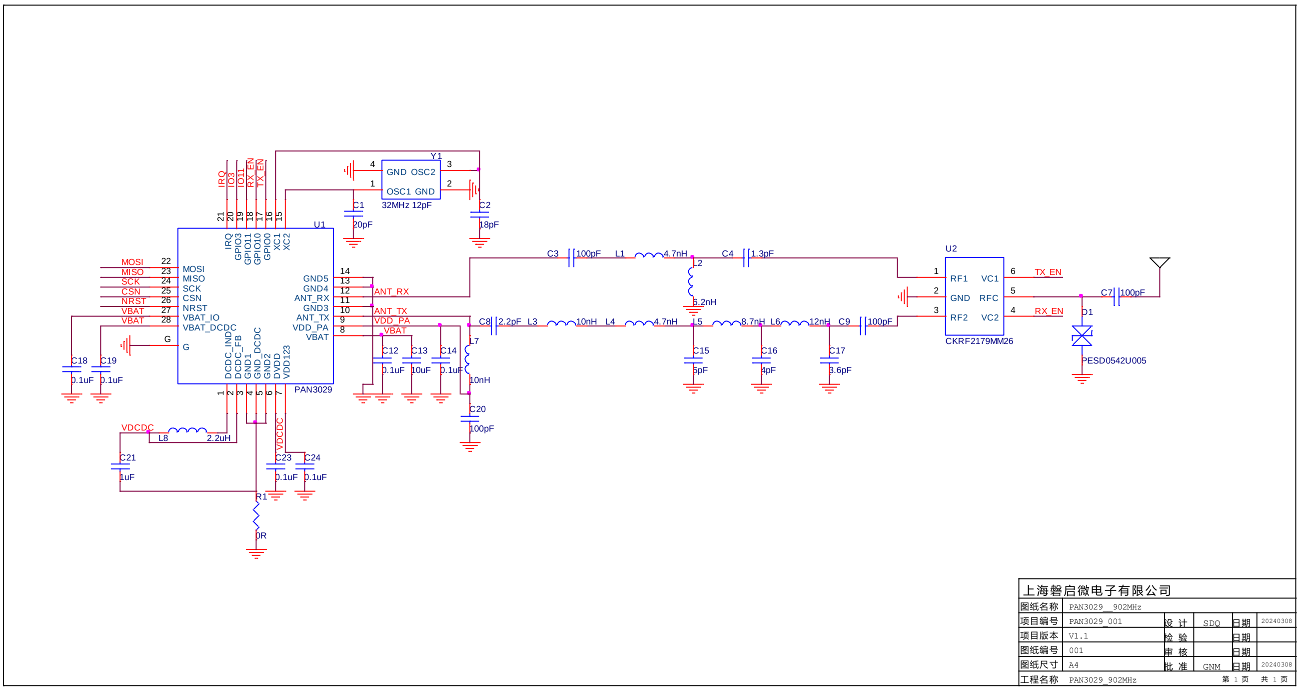


Figure 13-6 915 MHz reference schematic

14 Package size

图 14-1 QFN28 封装图

表 14-1 QFN28 封装尺寸

Dimension	Min (mm)	Typical (mm)	Max (mm)
A	0.70	0.75	0.80
A1	-	0.02	0.05
A2	0.203 REF		
b	0.15	0.20	0.25
D	3.90	4.00	4.10
D2	2.55	2.65	2.75
E	3.90	4.00	4.10
E2	2.55	2.65	2.75
e	0.40 BSC		
K	0.225	0.275	0.325
L	0.35	0.40	0.45
h	0.30	0.35	0.40
Ne	2.40 BSC		
Nd	2.40 BSC		

15 Precautions

1) This product is a CMOS device. Please pay attention to anti-static during storage, transportation and use.

2) The device must be well grounded when in use.

3) The reflow temperature cannot exceed 260℃.

The lead-free reflow process curve is as follows:

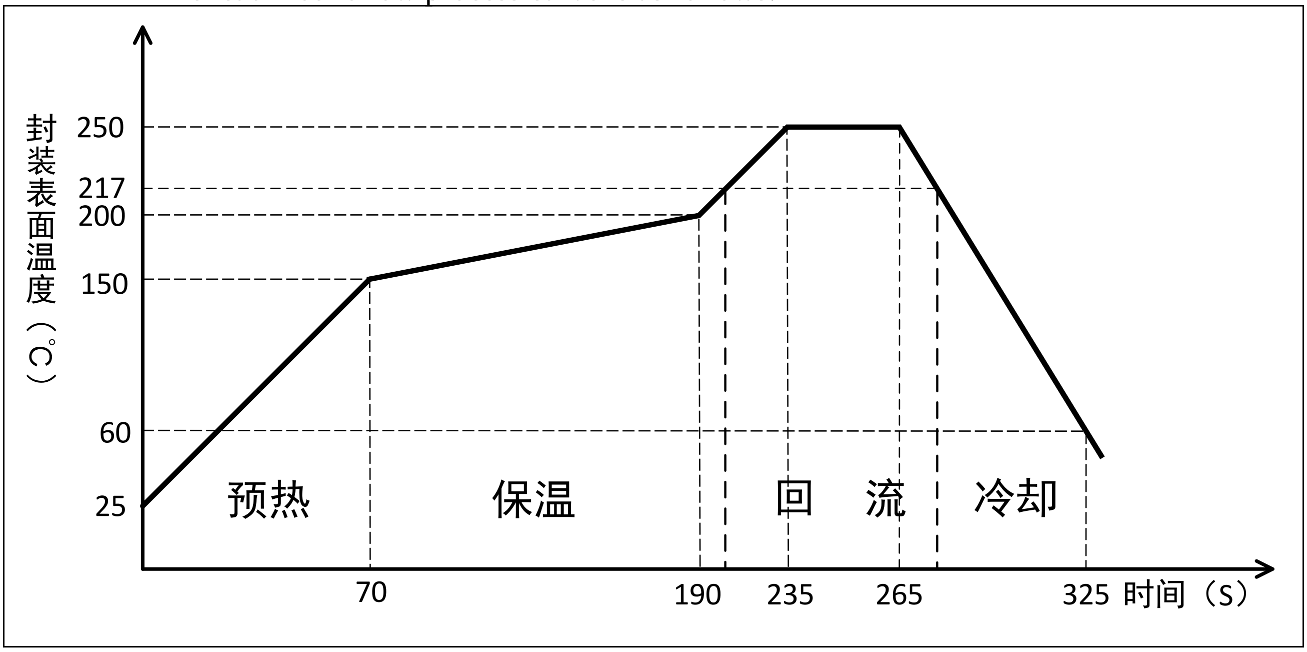


Figure 15-1 Reflow process curve

16 Storage conditions

1) Products are stored in sealed packages: up to 12 months at a temperature less than 30°C and a humidity less than 90%.

2) After the packaging bag is opened, the components must be used in the reflow process or other high-temperature processes and must meet the following requirements:

a) Completed within 72 hours and the factory environment is less than 30°C≤60%RH;

b) Stored in a 10%RH environment;

c) Baked at 125°C for 24 hours before use to remove internal moisture.

3) MSL (Packaging Moisture Sensitivity): Level 3 (determined according to IPC/JEDEC J-STD-020).

Abbreviations

ADC	Analog-to-digital converter
CAD	Channel activity detection
Chirp	Linear Frequency Modulation
CRC	Cyclic Redundancy Check
CSN	SPI Chip select signal
DAC	Digital to Analog Converter
DCDC	DC Converter
FIFO	First In First Out
GPIO	General input and output
IRQ	Interrupt Request
LDO	Low Dropout Linear Regulator
LPF	Low pass filter
MAC	Media Access Control Layer
MCU	Microcontroller unit
Mixer	Mixer
Modem	Modem
OSC	Oscillator
PA	Power Amplifier
RF	Radio Frequency
PLL	Phase-locked loop
PMU	Power Management Unit
POR	Power-On Reset
RAM	Random Access Memory
RSSI	Received signal strength indicator
SCK	SPI Clock signal
SF	Spreading Factor
SPI	Serial Peripheral Interface
STB	Standby mode
Sync	Synchronous
VCO	Voltage Controlled Oscillator

Revision History

Version	Date	Content
V1.0	2023.05	初版
V1.1	2023.06	增加包装方式、参考原理图，优化引脚定义中 pin1 和 pin2 名称，功能未变
V1.2	2023.07	增加 PAN3060 系列、电气特性等
V1.3	2023.08	更新参考原理图，补充工作电压描述
V1.4	2023.09	补充不同频段的参考原理图，补充不同频段的电气特性
V1.5	2023.10	更新参考原理图
V1.6	2023.12	更新基本操作配置中 SYNCWORD 寄存器的描述，更新参考原理图
V1.7	2024.03	更新参考原理图和电气特性。

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